

MODULAR KOSZUL DUALITY AND THE SHADOW TOWER: ALGEBRAICITY, DEPTH CLASSIFICATION, AND THE GRAVITATIONAL COPRODUCT

RAEEZ LORGAT

ABSTRACT. Let $\mathcal{A}$ be a chirally Koszul vertex algebra with modular characteristic κ , cubic shadow α , and quartic shadow S_4 . We prove four results. First, the weighted shadow generating function $H(t) = \sum_{r \geq 2} r S_r t^r$ satisfies $H(t)^2 = t^4 Q(t)$ where $Q(t) = 4\kappa^2 + 12\kappa\alpha t + (9\alpha^2 + 16\kappa S_4) t^2$ is the *shadow metric*: the tower is algebraic of degree 2, and every S_r for $r \geq 5$ is determined by the three seeds (κ, α, S_4) . Second, the critical discriminant $\Delta = 8\kappa S_4$ partitions the standard landscape into four depth classes: Gaussian ($r_{\max} = 2$), Lie ($r_{\max} = 3$), contact ($r_{\max} = 4$), and mixed ($r_{\max} = \infty$). Third, the genus-2 free energy is $F_2 = 7\kappa/5760$ for single-generator algebras; the first multi-channel computation for $\mathcal{W}_3$ exhibits a cross-channel correction $\delta F_2 = (c+204)/(16c)$, resolving the multi-generator universality question negatively. Fourth, the gravitational coproduct obtained by BRST reduction from $\widehat{\mathfrak{sl}}_2$ to Virasoro is strictly primitive at all degrees; the mechanism applies to all principal Drinfeld–Sokolov reductions.

Remark 0.1 (Relationship to companion papers). The c -independent identity $S_3(\text{Vir}_c) = 2$ is now proved as a theorem; see [16]. The seven-face programme connecting the shadow tower to Gaudin models, Sklyanin brackets, and GZ26 commuting Hamiltonians is developed in [17].

1. INTRODUCTION

1.1. The question. Classical Koszul duality is a phenomenon of graded algebras over a field. A quadratic algebra $\mathcal{A}$ determines a dual coalgebra $\mathcal{A}^!$; the bar construction $B(\mathcal{A})$ mediates between them; and the Koszul property (bar cohomology concentrated in bar degree 1) is equivalent to the existence of a linear resolution. The theory is the subject of Priddy [15], Beilinson–Ginzburg–Soergel [2], and Loday–Vallette [13].

On an algebraic curve X , the generators of the algebra become sections of a $\mathcal{D}_X$ -module, the quadratic relations become operator product expansions with meromorphic singularities, and the bar differential becomes an integral transform whose kernel is the logarithmic 1-form $\eta_{ij} = d \log(z_i - z_j)$ on the Fulton–MacPherson compactification $\overline{C}_n(X)$ of the configuration space. The resulting object, the *chiral bar complex* $\bar{B}^{\text{ch}}(\mathcal{A})$, is a factorization coalgebra on the Ran space $\text{Ran}(X)$ [1, 4, 6].

At genus 0, the classical theory embeds via restriction to the formal disk (the embedding requires specifying a retraction $\mathbb{A}^1 \rightarrow \text{pt}$ and its homotopy transfer; the E_1/E_∞ bar distinction, the Arnol’d algebra of $\text{Conf}_n(\mathbb{A}^1)$, and the Fulton–MacPherson compactifications are already absent over a bare point). At genus $g \geq 1$, the topology of the curve forces further structure: the bar differential acquires curvature $d_{\text{fb}}^2 = \kappa(\mathcal{A}) \cdot \omega_g$, where $\omega_g = c_1(\mathbb{E})$ is the Hodge class on $\overline{\mathcal{M}}_g$ and the scalar $\kappa(\mathcal{A}) \in \mathbb{Q}(c)$ is an intrinsic invariant of the algebra called the *modular characteristic*. This curvature has no analogue in the classical setting; its geometric origin is the nontriviality of the Hodge bundle over $\overline{\mathcal{M}}_g$.

Date: April 26, 2026.

2020 Mathematics Subject Classification. 17B69, 14H10, 18M70, 81T40.

Key words and phrases. Chiral algebras, bar construction, Koszul duality, moduli of curves, vertex algebras, Maurer–Cartan, W-algebras, gravitational coproduct.

The present paper addresses a single question: *what invariants does the chiral bar construction produce, beyond the modular characteristic?*

1.2. The answer: the shadow obstruction tower. The bar complex on a curve of genus g with n marked points defines a pronilpotent dg Lie algebra $\mathfrak{g}_{\mathcal{A}}^{\text{mod}}$ (the modular convolution algebra), whose differential encodes graph sums over stable curves. The boundary-of-boundary relation $\partial^2 = 0$ on $\overline{\mathcal{M}}_{g,n}$ implies $D^2 = 0$ on $\mathfrak{g}_{\mathcal{A}}^{\text{mod}}$, and the element $\Theta_{\mathcal{A}} := D_{\mathcal{A}} - d_0$ is therefore a Maurer–Cartan element: $D\Theta_{\mathcal{A}} + \frac{1}{2}[\Theta_{\mathcal{A}}, \Theta_{\mathcal{A}}] = 0$.

The MC element $\Theta_{\mathcal{A}}$ encodes the full genus expansion. Its finite-order projections $\Theta_{\mathcal{A}}^{\leq r}$, obtained by truncating the degree filtration at level r , form the *shadow obstruction tower*. At degree 2, the tower produces the modular characteristic κ . At degree 3, a cubic shadow α (the structure constant of the transferred L_{∞} bracket ℓ_3). At degree 4, a quartic shadow S_4 (the first genuinely nonlinear invariant, measuring the contact term in the four-point collision residue).

The obstruction to truncating $\Theta_{\mathcal{A}}$ at finite degree is measured by classes $o_{r+1}(\mathcal{A}) \in H^2(\mathcal{A}_{r+1,0}^{\text{sh}})$ in the shadow algebra $\mathcal{A}^{\text{sh}} := H_{\bullet}(\text{Def}_{\text{cyc}}^{\text{mod}}(\mathcal{A}))$. When $o_{r+1} = 0$, the tower stops; when $o_{r+1} \neq 0$, the next shadow coefficient S_{r+1} is forced. The question is how far this process continues and what controls it.

1.3. The shadow metric and algebraicity. A single quadratic polynomial controls the entire shadow obstruction tower (Theorem 4.1, parts (i)–(ii)).

Restrict to a one-dimensional primary slice L of the cyclic deformation complex, spanned by a single primary field η of definite conformal weight. On L , the MC recursion reduces to a scalar recursion for the shadow coefficients S_r , and the weighted generating function $H(t) = \sum_{r \geq 2} r S_r t^r$ satisfies

$$H(t)^2 = t^4 Q(t),$$

where the *shadow metric*

$$Q(t) = 4\kappa^2 + 12\kappa\alpha t + (9\alpha^2 + 16\kappa S_4) t^2$$

is a quadratic polynomial determined by the three seed invariants (κ, α, S_4) . The generating function is algebraic of degree 2: $H(t) = t^2 \sqrt{Q(t)}$, and every shadow coefficient S_r for $r \geq 5$ follows from the binomial expansion of $\sqrt{Q}$.

The proof converts the Maurer–Cartan recursion into a convolution identity for the Taylor coefficients of $F(t) = H(t)/t^2$, showing $F^2 = Q$. Orders $m = 0, 1, 2$ reproduce the definitions of κ, α, S_4 ; orders $m \geq 3$ are the recursion.

This is the *intrinsic quartic principle*: three numbers determine an infinite sequence of invariants. The analogue is the Weierstrass equation: two parameters (g_2, g_3) determine all Laurent coefficients of $\wp$, because $\wp$ satisfies an algebraic equation of degree 2 in $\wp'$. Here three parameters (κ, α, S_4) determine all S_r , because H satisfies an algebraic equation of degree 2.

1.4. The four-class classification. Write $Q(t) = (2\kappa + 3\alpha t)^2 + 2\Delta t^2$ for the *Gaussian decomposition*, where $\Delta := 8\kappa S_4$ is the *critical discriminant*. The Gaussian envelope $(2\kappa + 3\alpha t)^2$ is the part of Q visible to the quadratic and cubic shadows alone; Δ measures the irreducible quartic contribution.

The critical discriminant controls a dichotomy.

When $\Delta = 0$, the shadow metric is a perfect square: $Q = (2\kappa + 3\alpha t)^2$, $H(t) = t^2(2\kappa + 3\alpha t)$ is polynomial, and the tower terminates. If $\alpha = 0$, only $S_2 = \kappa$ survives (depth 2); if $\alpha \neq 0$, then S_2 and S_3 survive (depth 3).

When $\Delta \neq 0$, the shadow metric is irreducible: $\sqrt{Q}$ is irrational, every Taylor coefficient beyond degree 1 carries a factor of Δ , and $S_r \neq 0$ for all $r \geq 4$ generically. The tower is infinite.

Every chirally Koszul algebra whose primary-line shadow data has been computed falls into one of four *shadow depth classes* (Theorem 4.1, part (iii)). The partition is exhaustive for the standard Lie-theoretic landscape (Table 1).

Class	Δ	$r_{\max}$	Prototype	Shadow data
G (Gaussian)	0	2	Heisenberg	$\kappa = k$
L (Lie)	0	3	Affine Kac–Moody	$\kappa, \alpha \neq 0$
C (Contact)	0*	4	$\beta\gamma$ system	$\kappa = 6\lambda^2 - 6\lambda + 1$
M (Mixed)	$\neq 0$	∞	Virasoro, $\mathcal{W}_N$	κ, α, S_4

* Class **C**: $\Delta = 0$ on the primary line; the quartic contact invariant lives on a charged stratum.

Class **G**: the bar complex is classically formal, the tower terminates at κ , and all F_g follow from κ alone. Class **L**: a single cubic correction (the Lie structure constant) appears, after which the Jacobi identity kills the tower. Class **C**: the quartic contact invariant S_4 is nonzero but lives on a separate stratum whose self-bracket exits the complex, terminating the tower one step later. Class **M**: the tower is infinite, F_g receives corrections at every degree, and the shadow coefficients S_r decay or grow at a rate governed by the *shadow growth rate*

$$\rho(\mathcal{A}) = \sqrt{\frac{9\alpha^2 + 16\kappa S_4}{4\kappa^2}}.$$

The growth rate equals the reciprocal of the branch-point modulus. When $\rho < 1$, the tower converges absolutely; when $\rho > 1$, it diverges, but $\sqrt{Q}$ extends analytically past the convergence radius. For Virasoro at central charge c :

$$\rho(\text{Vir}_c) = \frac{1}{c} \sqrt{\frac{180c + 872}{5c + 22}},$$

with a critical central charge $c_\star \approx 6.125$ (the unique positive root of $5c^3 + 22c^2 - 180c - 872 = 0$) separating the convergent and divergent regimes. At the self-dual point $c = 13$ (where the Koszul involution $\text{Vir}_c \mapsto \text{Vir}_{26-c}$ has a fixed point, so Vir_{13} is self-dual under uncurved quadratic Koszul duality): $\rho \approx 0.467$.

Classical Koszul duality over a field embeds into the genus-0, degree-2, $\Delta = 0$ stratum (the Gaussian case where the tower terminates immediately) via restriction to the formal disk; the embedding is not an equivalence, since the configuration-space geometry and the E_1/E_∞ distinction are already absent over a bare point. Everything beyond class **G** is new.

1.5. The shadow connection and Koszul monodromy. The shadow metric equips each primary line with a logarithmic connection

$$\nabla^{\text{sh}} = d - \frac{Q'(t)}{2Q(t)} dt,$$

with regular singularities at the zeros of Q and residue $\frac{1}{2}$. The monodromy around each branch point is -1 . This is the *Koszul sign*: the square root $\sqrt{Q}$ changes sign around a simple zero, and the shadow obstruction tower inherits this sign change through the algebraic relation $H = t^2 \sqrt{Q}$.

Under Koszul duality $\mathcal{A} \mapsto \mathcal{A}^!$, the shadow metric transforms by the Verdier involution. For the Virasoro family $\text{Vir}_c^! = \text{Vir}_{26-c}$, the discriminants satisfy

$$\Delta(\text{Vir}_c) + \Delta(\text{Vir}_{26-c}) = \frac{6960}{(5c + 22)(152 - 5c)},$$

a Virasoro-specific identity whose numerator is $6960 = 2^4 \cdot 3 \cdot 5 \cdot 29$. The dual singularities $c = -22/5$ and $c = 152/5$ sum to 26: they are Koszul partners. Other families (affine Kac–Moody, $\mathcal{W}_N$ with $N \geq 3$) produce a different numerator.

1.6. Genus-2 free energies. The shadow obstruction tower produces invariants of moduli spaces through graph sums over stable curves. The genus- g free energy $F_g(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g^{\text{FP}}$ holds for single-generator (uniform-weight) chirally Koszul algebras at all genera, and unconditionally at genus 1 for all families. Here $\lambda_g^{\text{FP}} = \int_{\overline{\mathcal{M}}_{g,1}} \psi_1^{2g-2} \lambda_g$ is the Faber–Pandharipande integral, with $\lambda_g = c_g(\mathbb{E})$ the top Chern class of the Hodge bundle and ψ_1 the cotangent line class.

At genus 2: $\lambda_2^{\text{FP}} = 7/5760$, and $F_2(\mathcal{A}) = \kappa \cdot 7/5760$ for every single-generator chirally Koszul algebra. Verification requires an explicit sum over the seven stable graphs of genus 2 with no marked points (smooth genus-2 curve, figure-eight, banana, dumbbell, theta, tadpole, and chain), weighted by automorphism factors and vertex contributions from the shadow obstruction tower.

For the $\mathcal{W}_3$ algebra, which has two generators T (weight 2) and W (weight 3) producing two primary lines, the genus-2 computation is a multi-channel graph sum: each edge of the stable graph carries a channel label (T or W), each vertex contributes a shadow coefficient on the corresponding primary line, and the propagator $P_T = 2/c$, $P_W = 3/c$ depends on the channel. The per-channel diagonal sum yields $F_2^{\text{scal}}(\mathcal{W}_3) = \kappa \cdot \lambda_2^{\text{FP}} = 7c/6912$, but the full multi-channel graph sum produces an additional cross-channel correction $\delta F_2 = (c + 204)/(16c)$. The multi-generator universality problem is resolved negatively: the cross-channel correction is not cancelled by CohFT R -matrix terms, so $F_g \neq \kappa \cdot \lambda_g^{\text{FP}}$ at genus ≥ 2 for multi-weight algebras. The correct decomposition is $F_g = \kappa \cdot \lambda_g^{\text{FP}} + \delta F_g^{\text{cross}}$.

The Maurer–Cartan equation on $\overline{\mathcal{M}}_{2,0}$ also produces a *planted-forest correction*

$$\delta_{\text{pf}}^{(2,0)} = \frac{S_3(10S_3 - \kappa)}{48},$$

measuring the contribution of codimension-2 boundary strata to the genus-2 amplitude. This correction vanishes for class **G** (where $S_3 = 0$). For classes **C** and **M**, it is generically nonzero. For the Virasoro algebra: $\delta_{\text{pf}}^{(2,0)} = -(c - 40)/48$.

1.7. The gravitational coproduct. The fourth result concerns BRST reduction. Drinfeld–Sokolov (DS) reduction takes affine Kac–Moody data to $\mathcal{W}$ -algebra data via a BRST complex with ghosts; the homological perturbation lemma (HPL) transfers the $\mathcal{A}_\infty$ products and coproduct from the affine dg-shifted Yangian to a transferred Yangian on the $\mathcal{W}$ -algebra sector.

We prove the result for $\widehat{\mathfrak{sl}}_2 \rightarrow \text{Vir}$. The ghost-number obstruction (the homotopy h has ghost degree -1 ; the projection $p \otimes p$ kills all ghost-number- (-1) elements) forces the transferred coproduct to be *strictly primitive at all degrees*. The mechanism applies to any principal DS reduction $\widehat{\mathfrak{g}} \rightarrow \mathcal{W}(\widehat{\mathfrak{g}})$, since the ghost-number grading is universal; we state the general case as a conjecture and prove it for $\widehat{\mathfrak{sl}}_2$:

$$\Delta_z^{\text{Vir}}(T) = \tau_z(T) \otimes 1 + 1 \otimes T, \quad \Delta_{z,n}^{\text{Vir}} = 0 \quad \text{for } n \geq 2.$$

The ghost-number mechanism kills all higher-degree coproduct corrections. The only nontrivial gravitational structure is the r -matrix twist $r^{\text{Vir}}(z) = (c/2)/z^3 + 2T/z$, encoding the braiding of line operators.

The physical content: in Chern–Simons theory, gluons split (the Clebsch–Gordan decomposition makes the coproduct nontrivial). In 3d gravity, gravitons do not split: the coproduct is primitive, and all interactions reside in the twist.

1.8. Relation to existing work. The bar construction for chiral algebras was introduced by Beilinson and Drinfeld [1] and developed in the factorization algebra framework by Francis and Gaitsgory [6]

and Costello and Gwilliam [4]. The modular characteristic κ appears implicitly in the work of Feigin and Fuchs on Hodge bundles of conformal blocks, and the formula $\kappa(\text{Vir}_c) = c/2$ can be extracted from Beilinson and Schechtman [3]. The invariants of moduli spaces produced by vertex algebras via conformal blocks are the subject of Faber and Pandharipande [7], Mumford [14], and the extensive literature on cohomological field theories following Kontsevich and Manin [10].

The quartic shadow S_4 , the critical discriminant Δ , the four-class partition, and the algebraicity theorem (Theorem 4.1) appear to be new. The connection between shadow depth and L_∞ non-formality, while consonant with the general philosophy of Kontsevich [11], has not been established previously in the chiral setting.

The Drinfeld–Sokolov reduction and its homological algebra are classical [5, 8, 12]; the gravitational coproduct primitivity result (Theorem 9.2) and its connection to the shadow obstruction tower appear to be new.

Remark 1.1 (Relationship to companion papers). The Virasoro spectral R-matrix $R(z) = z^{2h} \exp(-(c/4)/z^2)$ and its connection to the Riccati shadow structure are developed in the companion paper *The Virasoro spectral R-matrix* (virasoro_r_matrix.tex).

2. THE SHADOW METRIC AND THE RICCATI EQUATION

2.1. The shadow obstruction tower. Let $\mathcal{A}$ be a chirally Koszul vertex algebra with modular cyclic deformation complex $\text{Def}_{\text{cyc}}^{\text{mod}}(\mathcal{A})$ and universal Maurer–Cartan element $\Theta_{\mathcal{A}} \in \text{MC}(\text{Def}_{\text{cyc}}^{\text{mod}}(\mathcal{A}))$. The complex carries a degree filtration $F^\bullet$ with $F^r = \bigoplus_{s \geq r} \text{Def}_{\text{cyc}}^{(s)}$, and the *shadow obstruction tower* is the sequence of finite-order truncations

$$\Theta_{\mathcal{A}}^{\leq 2}, \Theta_{\mathcal{A}}^{\leq 3}, \Theta_{\mathcal{A}}^{\leq 4}, \dots$$

where $\Theta_{\mathcal{A}}^{\leq r}$ solves the Maurer–Cartan equation in the quotient $\mathfrak{g}_{\mathcal{A}}^{\text{amb}}/F^{r+1}$. The degree- r projection $\text{Sh}_r(\mathcal{A})$ is the r -th *shadow* of $\mathcal{A}$. On a one-dimensional slice of the deformation complex, Sh_r is determined by a single scalar S_r ; the MC equation projected to degree r gives the *all-degree master equation*

$$\nabla_H(\text{Sh}_r) + \mathfrak{o}^{(r)} = 0, \quad (2.1)$$

where ∇_H is the linearized operator and $\mathfrak{o}^{(r)}$ is the obstruction from lower degrees.

2.2. Primary lines and the shadow metric.

Definition 2.1 (Primary line). A *primary line* in $\text{Def}_{\text{cyc}}^{\text{mod}}(\mathcal{A})$ is a one-dimensional subspace $L = \mathbf{k} \cdot \eta$ spanned by a single primary field η of definite conformal weight h_η . For a chiral algebra with generators $\phi_1, \dots, \phi_N$ of weights $h_1, \dots, h_N$, each generator spans a primary line; the full first-order deformation space is $\bigoplus_i L_i$.

Let L be a primary line with coordinate x , and write $\text{Sh}_r|_L = S_r x^r$ for the shadow data restricted to L . Set

$$\kappa := S_2, \quad \alpha := S_3.$$

The scalar κ is the modular characteristic (curvature) of $\mathcal{A}$ on L ; the scalar α is the cubic shadow.

Definition 2.2 (Shadow metric). Define the *weighted initial data* $a_0 := 2\kappa$, $a_1 := 3\alpha$, $a_2 := 4S_4$. The *shadow metric* (shadow quadratic form) on L is the polynomial

$$Q_L(t) := a_0^2 + 2a_0 a_1 t + (a_1^2 + 2a_0 a_2) t^2 = 4\kappa^2 + 12\kappa\alpha t + (9\alpha^2 + 16\kappa S_4) t^2 \in k(c)[t]. \quad (2.2)$$

The *critical discriminant* of L is

$$\Delta := 8\kappa S_4 = a_0 a_2.$$

Remark 2.3. The shadow metric Q_L is a quadratic polynomial in t , determined by three OPE invariants: κ, α, S_4 . The term “metric” refers to Q_L ’s role as the quadratic form governing the tower’s geometry (branch points, monodromy, convergence radius), not to a Riemannian metric.

2.3. The recursion on a primary line. On a primary line L , the all-degree master equation (2.1) reduces to a recursion among the scalars S_r .

Proposition 2.4 (Single-line recursion). *On a primary line L with propagator $P := 1/\kappa$, the MC equation at degree $r \geq 5$ is*

$$S_r = -\frac{P}{2r} \sum_{\substack{j+k=r+2 \\ 3 \leq j \leq k}} c_{jk} j k S_j S_k, \quad c_{jk} = \begin{cases} 1 & \text{if } j < k, \\ 1/2 & \text{if } j = k. \end{cases} \quad (2.3)$$

This recursion is well-defined: since $j \geq 3$ and $j + k = r + 2$, every term on the right has $k = r + 2 - j \leq r - 1$, so S_r is determined by $\{S_2, S_3, S_4, \dots, S_{r-1}\}$. The degree-4 shadow S_4 is an intrinsic OPE invariant (the quartic contact term in the four-point collision residue), not determined by the recursion; it serves as initial data alongside $\kappa = S_2$ and $\alpha = S_3$.

Proof. The Lie bracket on the cyclic deformation complex, restricted to L , pairs $\text{Sh}_j = S_j x^j$ and $\text{Sh}_k = S_k x^k$ to produce a term of degree $j + k - 2$ with coefficient $jk P S_j S_k$. The eigenvalue identity $\nabla_H(x^r) = 2r x^r$ separates the MC equation at degree r into the linear term $2r S_r$ on the left and the obstruction sum on the right, giving the stated recursion. The symmetry factor c_{jk} accounts for the factor $\frac{1}{2}$ in $\frac{1}{2}[\Theta, \Theta]$ when $j = k$. $\square$

2.4. The weighted shadow generating function.

Definition 2.5 (Weighted shadow generating function). *The weighted shadow generating function on L is*

$$H(t) := \sum_{r \geq 2} r S_r t^r.$$

Setting $a_n := (n + 2) S_{n+2}$ for $n \geq 0$, this becomes $H(t) = \sum_{n \geq 0} a_n t^{n+2}$, with initial values $a_0 = 2\kappa$, $a_1 = 3\alpha$, $a_2 = 4S_4$.

The weighted generating function H packages the recursion (2.3) as a convolution identity. Set $F(t) := H(t)/t^2 = \sum_{n \geq 0} a_n t^n$. The MC recursion forces every Cauchy coefficient of F^2 beyond degree 2 to vanish, so F^2 is a polynomial of degree 2.

3. ALGEBRAICITY AND THE DEPTH CLASSIFICATION

3.1. The Riccati algebraicity theorem.

Theorem 3.1 (Riccati algebraicity). *Let L be a primary line with shadow data S_r , shadow metric Q_L (Definition 2.2), and weighted shadow generating function H (Definition 2.5). Then the all-degree master equation on L is equivalent to the algebraic relation*

$$H(t)^2 = t^4 Q_L(t). \quad (3.1)$$

The shadow generating function is therefore algebraic of degree 2 over $k(\kappa, \alpha, S_4)(t)$:

$$H(t) = t^2 \sqrt{Q_L(t)}.$$

Proof. Set $F(t) := H(t)/t^2 = \sum_{n \geq 0} a_n t^n$ where $a_n = (n + 2) S_{n+2}$. The claim (3.1) is equivalent to $F(t)^2 = Q_L(t)$. We verify this by computing the Cauchy product $F^2 = \sum_{m \geq 0} (\sum_{i+j=m} a_i a_j) t^m$ and showing it agrees with Q_L at all orders.

Orders 0, 1, 2 (initial data). At $m = 0$: $a_0^2 = (2\kappa)^2 = 4\kappa^2$. At $m = 1$: $2 a_0 a_1 = 2 \cdot 2\kappa \cdot 3\alpha = 12\kappa\alpha$. At $m = 2$: $a_1^2 + 2 a_0 a_2 = 9\alpha^2 + 2 \cdot 2\kappa \cdot 4S_4 = 9\alpha^2 + 16\kappa S_4$. These are exactly the coefficients of $Q_L(t)$ in (2.2).

Orders $m \geq 3$ (the MC recursion forces vanishing). Since Q_L has degree 2 in t , the identity $F^2 = Q_L$ requires

$$\sum_{i+j=m} a_i a_j = 0 \quad \text{for all } m \geq 3. \quad (3.2)$$

We show that this is exactly the MC recursion (2.3) at degree $r = m + 2$.

The sum $\sum_{i+j=m} a_i a_j$ with $a_i = (i+2) S_{i+2}$ and $a_j = (j+2) S_{j+2}$ equals $\sum_{p+q=m+4, p,q \geq 2} p q S_p S_q$, where $p = i+2$, $q = j+2$. Setting $r = m+2$, this becomes $\sum_{p+q=r+2, p,q \geq 2} p q S_p S_q$.

Isolating the terms with $p = 2$ or $q = 2$ (which contribute 2κ from $S_2 = \kappa$):

$$\begin{aligned} \sum_{p+q=r+2, p,q \geq 2} p q S_p S_q &= 2 \cdot 2r \cdot \kappa S_r + \sum_{\substack{p+q=r+2 \\ p,q \geq 3}} p q S_p S_q \\ &= 4r\kappa S_r + 2 \sum_{\substack{j+k=r+2 \\ 3 \leq j \leq k}} c_{jk} j k S_j S_k, \end{aligned}$$

where the factor 2 in front of the restricted sum accounts for $S_p S_q + S_q S_p$ when $p \neq q$, and c_{jk} corrects for the diagonal. Substituting the recursion (2.3):

$$2 \sum_{\substack{j+k=r+2 \\ 3 \leq j \leq k}} c_{jk} j k S_j S_k = 2 \cdot (-2r\kappa S_r) = -4r\kappa S_r.$$

Hence $\sum_{p+q=r+2} p q S_p S_q = 4r\kappa S_r - 4r\kappa S_r = 0$, which is (3.2).

Conversely, the vanishing (3.2) at order $m \geq 3$ implies the recursion (2.3) at $r = m+2$ by solving for S_r (using $\kappa \neq 0$).

The equivalence of (3.1) with the full MC recursion is therefore established at all degrees. Taking positive square roots (the branch with $F(0) = a_0 = 2\kappa > 0$) gives $H(t) = t^2 F(t) = t^2 \sqrt{Q_L(t)}$. $\square$

Remark 3.2 (Riccati ODE reformulation). Set $U(t) := \sum_{r \geq 3} S_r t^r$ (the nonlinear part, removing Sh_2). The algebraic relation (3.1) is equivalent to the Riccati-type first-order ODE

$$2t U'(t) + \frac{P}{2} (U'(t))^2 = R_3 t^3 + R_4 t^4,$$

where $R_3 = 6\alpha$ and $R_4 = (9\alpha^2 + 2\Delta)/(2\kappa)$ encode the cubic and quartic inputs. The left-hand side is quadratic in U' : this is the precise sense in which the shadow obstruction tower is algebraic of degree 2.

3.2. Gaussian decomposition.

Corollary 3.3 (Gaussian decomposition). *The shadow metric decomposes as*

$$Q_L(t) = (2\kappa + 3\alpha t)^2 + 2\Delta t^2,$$

where $\Delta = 8\kappa S_4$ is the critical discriminant. The first term $(2\kappa + 3\alpha t)^2$ is the Gaussian envelope: the shadow metric of the truncated tower with $S_r = 0$ for $r \geq 4$. The second term $2\Delta t^2$ is the interaction correction, measuring the quartic OPE contribution not captured by the cubic data alone.

Proof. $(2\kappa + 3\alpha t)^2 = 4\kappa^2 + 12\kappa\alpha t + 9\alpha^2 t^2$. Adding $2\Delta t^2 = 16\kappa S_4 t^2$ recovers the coefficient $9\alpha^2 + 16\kappa S_4$ of t^2 in $Q_L(t)$. $\square$

3.3. Intrinsic quartic principle.

Corollary 3.4 (Intrinsic quartic principle). *On a primary line L , the shadow obstruction tower has intrinsic OPE data at degrees 2, 3, 4 only. All shadow coefficients S_r for $r \geq 5$ are determined by (κ, α, S_4) alone, via the closed form*

$$S_r = \frac{1}{r} [t^{r-2}] \sqrt{Q_L(t)} \quad (r \geq 2).$$

Proof. $Q_L(t) = 4\kappa^2 + 12\kappa\alpha t + (9\alpha^2 + 16\kappa S_4) t^2$ depends on exactly (κ, α, S_4) . Since $H(t) = t^2 \sqrt{Q_L(t)} = \sum_{r \geq 2} r S_r t^r$ by Theorem 3.1, comparing coefficients gives $r S_r = [t^{r-2}] \sqrt{Q_L}$. $\square$

3.4. The shadow connection.

Proposition 3.5 (Shadow connection). *The shadow metric defines a logarithmic connection on $\mathcal{O}_L$ over $L \setminus \{Q_L = 0\}$:*

$$\nabla^{\text{sh}} = d - \omega, \quad \omega := \frac{Q'_L(t)}{2Q_L(t)} dt = d(\log \sqrt{Q_L}).$$

- (i) *The connection is flat: $(\nabla^{\text{sh}})^2 = 0$.*
- (ii) *At each simple zero t_0 of Q_L , the residue of ω is $\frac{1}{2}$.*
- (iii) *The monodromy is $\exp(2\pi i \cdot \frac{1}{2}) = -1$: the standard monodromy of a square root around a simple branch point.*
- (iv) *The flat section with $\Phi(0) = 1$ is $\Phi(t) = \sqrt{Q_L(t)/Q_L(0)} = \sqrt{Q_L(t)/(2\kappa)}$, and the shadow generating function is $H(t) = 2\kappa t^2 \Phi(t)$: the parallel transport of the curvature along the degree parameter, weighted by t^2 .*

Proof. Flatness is automatic: ω is a closed 1-form on a 1-dimensional base. At a simple zero t_0 of Q_L : $Q_L(t) \approx Q'_L(t_0)(t - t_0)$, so $\omega \approx \frac{1}{2(t-t_0)} dt$, giving residue $\frac{1}{2}$. The monodromy exponentiation is immediate. For (iv): $\nabla^{\text{sh}}\Phi = 0$ since $d\Phi = \omega\Phi$ by construction, and the evaluation at $t = 0$ gives $\Phi(0) = \sqrt{Q_L(0)/(2\kappa)} = 2\kappa/(2\kappa) = 1$. $\square$

Remark 3.6. The shadow generating function $H(t) = 2\kappa t^2 \Phi(t)$ is *not* a flat section of ∇^{sh} : the factor t^2 contributes $\nabla^{\text{sh}}(H) \neq 0$. The flat section is $\Phi(t) = \sqrt{Q_L(t)/(2\kappa)}$; the shadow obstruction tower is the Taylor expansion of $t^2 \cdot \Phi(t)$, reflecting the degree offset (the tower starts at degree 2, not degree 0).

3.5. The single-line dichotomy.

Theorem 3.7 (Single-line dichotomy). *Let $L, S_r, \alpha, \Delta, Q_L$ be as above, and define the shadow depth $r_{\max}|_L := \max\{r : S_r \neq 0\}$ (with $r_{\max} = \infty$ if $S_r \neq 0$ for infinitely many r). Then $r_{\max}|_L \in \{2, 3, \infty\}$, classified by the shadow metric:*

- (i) **Class G** ($\Delta = 0, \alpha = 0$). $Q_L = (2\kappa)^2$ is constant. $H(t) = 2\kappa t^2$. Only S_2 survives: $r_{\max} = 2$.
- (ii) **Class L** ($\Delta = 0, \alpha \neq 0$). $Q_L = (2\kappa + 3\alpha t)^2$ is a perfect square. $H(t) = t^2(2\kappa + 3\alpha t)$. S_2 and S_3 survive: $r_{\max} = 3$.
- (iii) **Class M** ($\Delta \neq 0$). Q_L is irreducible over $k(c)$; $\sqrt{Q_L}$ is irrational; $S_r \neq 0$ for all $r \geq 4$ generically: $r_{\max} = \infty$.

For $r \geq 4$,

$$S_r = \Delta \cdot R_r, \quad R_r \in \mathbb{Q}(\alpha, \kappa, S_4). \quad (3.3)$$

Proof. The classification follows from Theorem 3.1 and the Gaussian decomposition (Corollary 3.3).

Case $\Delta = 0$. Then $Q_L = (2\kappa + 3\alpha t)^2$ is a perfect square, so $\sqrt{Q_L} = 2\kappa + 3\alpha t$ is polynomial and $H(t) = t^2(2\kappa + 3\alpha t)$ terminates. If $\alpha = 0$: $H(t) = 2\kappa t^2$, giving $r_{\max} = 2$. If $\alpha \neq 0$: $H(t) = 2\kappa t^2 + 3\alpha t^3$, giving $r_{\max} = 3$.

Case $\Delta \neq 0$. The discriminant of Q_L as a polynomial in t is

$$(12\kappa\alpha)^2 - 4 \cdot 4\kappa^2 \cdot (9\alpha^2 + 16\kappa S_4) = -32\kappa^2\Delta \neq 0,$$

so Q_L is irreducible over $k(c)$. The Taylor expansion of $\sqrt{Q_L}$ around $t = 0$ has nonzero coefficients at all orders generically, hence $r_{\max} = \infty$.

Universal factorization. The binomial expansion of $\sqrt{Q_L}$ around the Gaussian envelope $(2\kappa + 3\alpha t)$ is

$$\sqrt{Q_L} = (2\kappa + 3\alpha t) \sqrt{1 + \frac{2\Delta t^2}{(2\kappa + 3\alpha t)^2}}.$$

Expanding $\sqrt{1+u}$ as $1 + \frac{1}{2}u - \frac{1}{8}u^2 + \dots$ with $u = 2\Delta t^2 / (2\kappa + 3\alpha t)^2$, every coefficient of t^n for $n \geq 2$ in $\sqrt{Q_L} - (2\kappa + 3\alpha t)$ carries a factor of Δ . Since $rS_r = [t^{r-2}] \sqrt{Q_L}$ for $r \geq 2$, and the Gaussian part $2\kappa + 3\alpha t$ contributes only at $r \leq 3$, the factorization $S_r = \Delta \cdot R_r$ for $r \geq 4$ follows. $\square$

Remark 3.8 (The contact class). The partition $\mathbf{G}/\mathbf{L}/\mathbf{C}/\mathbf{M}$ of chirally Koszul algebras includes a fourth class $\mathbf{C}$ (contact, $r_{\max} = 4$), realized by the $\beta\gamma$ system. This class escapes the single-line dichotomy by *stratum separation*: the quartic contact invariant lives on a charged stratum of $\text{Def}_{\text{cyc}}^{\text{mod}}$ where $\kappa = 0$, and the shadow metric requires $\kappa \neq 0$. On the charged stratum, the self-bracket maps to a zero-dimensional target by rank-one rigidity, so the quartic pump does not activate and the tower terminates at $r = 4$. The algebraic shadow depth $d_{\text{alg}} \in \{0, 1, 2, \infty\}$ exhausts the four classes: $\mathbf{G}$ ($d_{\text{alg}} = 0$), $\mathbf{L}$ ($d_{\text{alg}} = 1$), $\mathbf{C}$ ($d_{\text{alg}} = 2$), $\mathbf{M}$ ($d_{\text{alg}} = \infty$).

3.6. The shadow growth rate. For class $\mathbf{M}$ algebras, the shadow obstruction tower is infinite but governed by a single analytic invariant: the growth rate of the coefficients S_r .

Definition 3.9 (Shadow growth rate). Let L be a primary line of class $\mathbf{M}$ ($\Delta \neq 0$), and write $Q_L(t) = q_0 + q_1t + q_2t^2$ for the shadow metric. The *shadow growth rate* of L is

$$\rho_L := \sqrt{\frac{q_2}{q_0}} = \frac{\sqrt{9\alpha^2 + 2\Delta}}{2|\kappa|}.$$

By Vieta's theorem, the zeros $t_0, \bar{t}_0$ of Q_L satisfy $|t_0|^2 = q_0/q_2$, so $\rho_L = 1/|t_0|$: the reciprocal of the branch-point modulus. For classes $\mathbf{G}$ and $\mathbf{L}$, set $\rho := 0$ by convention.

Theorem 3.10 (Shadow asymptotics). Let L be a primary line of class $\mathbf{M}$ with shadow growth rate ρ_L .

(i) Asymptotic formula.

$$S_r \sim A(\mathcal{A}) \rho_L^r r^{-5/2} \cos(r\theta + \varphi) \quad (r \rightarrow \infty),$$

where $\theta := -\arg(t_0)$ is the oscillation phase, $A(\mathcal{A}) > 0$, and φ is computable from the branch-point residue of $\sqrt{Q_L}$. The exponent $-5/2$ arises from $S_r = [t^{r-2}] \sqrt{Q_L}/r$ combined with the $n^{-3/2}$ transfer law for square-root singularities.

(ii) Unit circle criterion. $\rho_L = 1$ if and only if $q_0 = q_2$, i.e.,

$$(2\kappa - 3\alpha)(2\kappa + 3\alpha) = 2\Delta.$$

Equivalently, the branch points of Q_L lie on the unit circle $|t| = 1$.

(iii) Convergence partition. $\rho_L < 1$: the shadow obstruction tower converges absolutely. $\rho_L > 1$: the shadow obstruction tower diverges, but the algebraic function $\sqrt{Q_L(t)}$ extends analytically past the convergence radius. $\rho_L = 1$: marginal case (the critical surface).

Proof. The function $\sqrt{Q_L(t)}$ has algebraic branch-cut singularities of order $\frac{1}{2}$ at each simple zero t_0 of Q_L . Since Q_L is quadratic with nonzero discriminant ($\Delta \neq 0$ implies $\text{disc}(Q_L) = -32\kappa^2\Delta \neq 0$), the two zeros $t_0, \bar{t}_0$ are distinct with $|t_0| = |\bar{t}_0| = \sqrt{q_0/q_2} = 1/\rho_L$.

The transfer theorem for algebraic singularities [9, Theorem VI.1] gives $[t^n]\sqrt{Q_L} \sim C_0 t_0^{-n} n^{-3/2}$ for each dominant singularity t_0 at distance $R = 1/\rho_L$ from the origin. Both branch points contribute (equal modulus), producing the cosine interference term from $2 \operatorname{Re}(C_0 t_0^{-(r-2)})$. Since $S_r = [t^{r-2}]\sqrt{Q_L}/r$ (Corollary 3.4), the extra factor $1/r$ gives the stated $r^{-5/2}$ exponent.

For (ii): $\rho_L = 1$ iff $q_2/q_0 = 1$. Writing $q_0 = 4\kappa^2$ and $q_2 = 9\alpha^2 + 2\Delta$, the condition $q_0 = q_2$ factors as $4\kappa^2 - 9\alpha^2 = 2\Delta$. Part (iii) is immediate from the definition of the convergence radius $R = 1/\rho_L$. $\square$

3.7. The depth decomposition.

Theorem 3.11 (Depth decomposition). *The shadow depth of a chirally Koszul algebra $\mathcal{A}$ decomposes as*

$$d(\mathcal{A}) = 1 + d_{\text{arith}}(\mathcal{A}) + d_{\text{alg}}(\mathcal{A}),$$

where:

- (i) d_{arith} counts the independent holomorphic Hecke eigenforms in the spectral decomposition of the completed shadow partition function $\widehat{Z}_{\mathcal{A}}^c$ on $\mathcal{M}_{1,1}$;
- (ii) $d_{\text{alg}} \in \{0, 1, 2, \infty\}$ is the algebraic depth, determined by the single-line dichotomy (Theorem 3.7) and stratum separation (Remark 3.8):

$$d_{\text{alg}} = \begin{cases} 0 & \text{class } \mathbf{G}, \\ 1 & \text{class } \mathbf{L}, \\ 2 & \text{class } \mathbf{C}, \\ \infty & \text{class } \mathbf{M}; \end{cases}$$

- (iii) depths $d \geq 5$ are purely arithmetic: they arise from d_{arith} , not from the A_∞ structure.

Proof. The algebraic depth d_{alg} is determined by the structure of the cyclic deformation complex and the MC recursion: the single-line dichotomy (Theorem 3.7) gives $d_{\text{alg}} \in \{0, 1, \infty\}$ on each primary line, and stratum separation (Remark 3.8) provides $d_{\text{alg}} = 2$ for the contact class. Since $d_{\text{alg}} \leq 2$ for finite cases ($\mathbf{G}/\mathbf{L}/\mathbf{C}$) and the base term is 1 (the curvature κ at degree 2), the maximum finite depth without arithmetic contributions is $1 + 0 + 2 = 3$; hence all depths $d \geq 5$ require $d_{\text{arith}} \geq 2$.

The arithmetic depth counts independent holomorphic Hecke eigenforms by decomposing $\widehat{Z}_{\mathcal{A}}^c$ via the Roelcke–Selberg spectral decomposition on $\mathcal{M}_{1,1}$. Each holomorphic eigenform of weight k contributes a critical line at $\operatorname{Re}(s) = k/2$; Maass cusp-form projections contribute no new critical lines. $\square$

Example 3.12 (Standard families). The shadow metric data for the standard families:

Family	κ	α	S_4	Δ	Class	ρ
Heisenberg $\mathcal{H}_k$	k	0	0	0	G	0
Affine $\widehat{\mathfrak{g}}_k$	$\frac{\dim \mathfrak{g}(k+h^\vee)}{2h^\vee}$	$\neq 0$	0	0	L	0
$\beta\gamma$ (weight λ)	$6\lambda^2 - 6\lambda + 1$	0*	0*	0*	C	0
Virasoro Vir_c	$c/2$	2	$\frac{10}{c(5c+22)}$	$\frac{40}{5c+22}$	M	$\frac{1}{c} \sqrt{\frac{180c+872}{5c+22}}$

* The table reports the weight-changing primary line, where $\kappa = S_3 = S_4 = 0$. The quartic contact invariant $Q^{\text{contact}} \neq 0$ lives on this charged stratum; the shadow metric requires $\kappa \neq 0$ and does not

apply. The class **C** designation ($r_{\max} = 4$) comes from rank-one rigidity, not the single-line dichotomy. The T-line (stress tensor) carries the Virasoro shadow data $(\kappa, \alpha, S_4) = (c/2, 2, 10/(c(5c+22)))$ with $c = 2(6\lambda^2 - 6\lambda + 1)$ and is class **M**.

For the Virasoro algebra, the shadow metric evaluates to

$$Q_{\text{Vir}}(t) = c^2 + 12c t + \frac{180c + 872}{5c + 22} t^2,$$

with Gaussian decomposition $Q_{\text{Vir}} = (c + 6t)^2 + \frac{80}{5c+22} t^2$. The critical central charge c^* where $\rho = 1$ is the unique positive real root of

$$5c^3 + 22c^2 - 180c - 872 = 0,$$

approximately $c^* \approx 6.12$. For $c > c^*$ the shadow tower converges; for $c < c^*$ it diverges. At the Koszul self-dual point $c = 13$: $\rho(\text{Vir}_{13}) \approx 0.467$.

Remark 3.13 (Large- c asymptotic of S_4). The Virasoro quartic shadow

$$S_4(\text{Vir}_c) = \frac{10}{c(5c + 22)}$$

has large- c asymptotic

$$S_4(\text{Vir}_c) = \frac{10}{5c^2} \left(1 - \frac{22}{5c} + O(c^{-2}) \right) = \frac{2}{c^2} + O(c^{-3}),$$

not $2/(5c^2)$; the factor of 5 in the denominator of $10/[c(5c + 22)]$ is absorbed when expanding $1/(5c + 22)$ at large c . Pairing this with $\kappa = c/2$ gives the critical discriminant $\Delta = 8\kappa S_4 = 40/(5c + 22) \rightarrow 8/c$ as $c \rightarrow \infty$, confirming class **M** mixing persists to leading order.

The identity $S_2(\text{Vir}_c) = \kappa = c/2$ is convention-independent. The occasional error $S_2 = c/12$ conflates the shadow invariant with the Vol II lambda-bracket divided-power coefficient $c/12 = (c/2)/3!$, which is a different normalisation; the Vol I–II bridge is $S_2^{\text{Vol I}} = 6 S_2^{\lambda}$.

The Riccati seed on Vir_c is $(r_o, S_{r_o}) = (3, 2)$: $S_3(\text{Vir}_c) = 2$ is the c -independent cubic shadow (Lorgat, *The Virasoro spectral R-matrix*), and the shadow-exponential base $C_A = 6$ on the non-logarithmic class **M** locus factorises as $6 = r_o \cdot S_{r_o} = 3 \cdot 2$.

Remark 3.14 (Spectral curve). The algebraic relation $H^2 = t^4 Q_L(t)$ defines, for each primary line L , a spectral curve $\Sigma_L := \{H^2 = t^4 Q_L(t)\} \subset \mathbb{A}_{t,H}^2$. Since Q_L is quadratic in t , Σ_L has arithmetic genus 0; the projection $(t, H) \mapsto t$ is a double cover ramified at the zeros of Q_L . The degeneration loci in parameter space are controlled by the critical discriminant: Σ_L acquires a node precisely when $\Delta = 0$. The shadow growth rate ρ_L is the reciprocal of the distance from $t = 0$ to the nearest ramification point.

4. MAIN RESULTS

Theorem 4.1 (Main theorem: algebraicity, classification, and genus-2 formula). *Let $\mathcal{A}$ be a chirally Koszul vertex algebra on a smooth projective curve X . Let $\kappa = \kappa(\mathcal{A})$ be the modular characteristic, and let α, S_4 be the cubic and quartic shadow invariants. Define the shadow metric*

$$Q(t) = 4\kappa^2 + 12\kappa\alpha t + (9\alpha^2 + 16\kappa S_4) t^2$$

and the critical discriminant $\Delta = 8\kappa S_4$.

(i) **Algebraicity.** *The weighted shadow generating function $H(t) = \sum_{r \geq 2} r S_r t^r$ satisfies*

$$H(t)^2 = t^4 Q(t).$$

Every shadow coefficient S_r for $r \geq 5$ is determined by (κ, α, S_4) via $S_r = \frac{1}{r} [t^{r-2}] \sqrt{Q(t)}$.

- (ii) **Shadow connection.** The shadow metric defines a logarithmic connection $\nabla^{\text{sh}} = d - Q'/(2Q) dt$ with residue $\frac{1}{2}$ at each zero of Q . Monodromy $= -1$ (Koszul sign).
- (iii) **Four-class partition.** Write $Q = (2\kappa + 3\alpha t)^2 + 2\Delta t^2$. The shadow depth $r_{\max}$ lies in $\{2, 3, 4, \infty\}$:
- Class **G**: $\Delta = 0, \alpha = 0, r_{\max} = 2$.
 - Class **L**: $\Delta = 0, \alpha \neq 0, r_{\max} = 3$.
 - Class **C**: $\Delta = 0$ on the primary line; quartic contact on a charged stratum, $r_{\max} = 4$.
 - Class **M**: $\Delta \neq 0, r_{\max} = \infty$.
- (iv) **Universal factorization.** For $r \geq 4$: $S_r = \Delta \cdot R_r$, where $R_r \in \mathbb{Q}(\alpha, \kappa, S_4)$ is independent of the algebra.
- (v) **Planted-forest formula.** The genus-2 planted-forest correction on a primary line is $\delta_{\text{pf}}^{(2,0)} = S_3(10S_3 - \kappa)/48$. It vanishes for class **G** (where $S_3 = 0$).

Proof. Parts (i)–(ii) are proved in Theorem 3.1 and Proposition 3.5, respectively. Part (iii) follows from Theorem 3.7. Part (iv) is the universal factorization in Theorem 3.7, equation (3.3). Part (v) is proved by explicit computation in Section 8. $\square$

Remark 4.2 (Relationship to companion papers). The three-invariants distinction $p_{\max} \neq k_{\max} \neq r_{\max}$ and the operator-order trichotomy are developed further in the companion paper *The classification trichotomy for chiral algebras* (classification_trichotomy.tex).

5. THE FOUR SHADOW CLASSES

The single-line dichotomy (Theorem 3.7) partitions the space of chirally Koszul algebras into four shadow classes, determined by two invariants of the shadow metric $Q_L(t) = (2\kappa + 3\alpha t)^2 + 2\Delta t^2$: the cubic coefficient $\alpha = S_3$ and the critical discriminant $\Delta = 8\kappa S_4$.

Definition 5.1 (Shadow classes). Let $\mathcal{A}$ be a chirally Koszul algebra with shadow metric Q_L on a one-dimensional primary line L . The *shadow class* of $\mathcal{A}$ along L is:

Class	α	Δ	$r_{\max}$	Q_L
G (Gaussian)	0	0	2	$(2\kappa)^2$
L (Lie/tree)	$\neq 0$	0	3	$(2\kappa + 3\alpha t)^2$
C (Contact)	0	0*	4	$(2\kappa)^2$ (primary line)
M (Mixed)	$\neq 0$	$\neq 0$	∞	irreducible quadratic

Here $r_{\max}$ is the *shadow depth*: the largest degree r for which the shadow coefficient $S_r \neq 0$. *Class C has $\Delta = 0$ on the primary line (where the single-line dichotomy applies), but the quartic contact invariant $Q^{\text{contact}} \neq 0$ lives on a charged stratum; see Definition 5.4.

The classification is exhaustive: every entry in Table 1 falls into exactly one class, and every class is realized by at least one standard family. The mechanism is algebraic: the shadow generating function $H(t) = t^2 \sqrt{Q_L(t)}$ is algebraic of degree 2 (Theorem 3.1), and its Taylor coefficients S_r are computed from the convolution recursion $H(t)^2 = t^4 Q_L(t)$, where $H(t) = \sum_{r \geq 2} S_r t^r$ and $S_r = a_{r-2}/r$ with a_n the Taylor coefficients of $\sqrt{Q_L}$. When Q_L is a perfect square ($\Delta = 0$), the square root terminates; when Q_L is irreducible ($\Delta \neq 0$), the Taylor expansion is infinite.

5.1. Class G: Gaussian ($r_{\max} = 2$).

Definition 5.2. A chirally Koszul algebra $\mathcal{A}$ is *Gaussian* if $S_r(\mathcal{A}) = 0$ for all $r \geq 3$. Equivalently, $\alpha = 0$ and $\Delta = 0$, so the shadow metric $Q_L(t) = 4\kappa^2$ is constant and the shadow obstruction tower terminates at the scalar curvature.

Examples. Heisenberg $\mathcal{H}_k$ with $\kappa = k$; lattice VOAs V_Λ with $\kappa = \text{rank}(\Lambda)$; free fermion ψ with $\kappa = 1/4$; bc ghosts with $\kappa = -(6\lambda^2 - 6\lambda + 1)$.

Characterization. Class G algebras are freely generated as chiral algebras. Their bar complex is quadratic (the differential has no higher-order terms), and the OPE on each primary line is abelian. The full obstruction is captured by a single scalar:

$$\Theta_{\mathcal{A}} = \kappa(\mathcal{A}) \cdot \eta \otimes \Lambda,$$

where η is the cyclic direction and $\Lambda \in H^*(\tilde{\mathcal{M}}_g)$ is the tautological class. All genus- g data is controlled by $F_g(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g^{\text{FP}}$ (uniform-weight).

5.2. Class L: Lie/tree ($r_{\max} = 3$).

Definition 5.3. A chirally Koszul algebra $\mathcal{A}$ is *Lie/tree* if $\alpha \neq 0$ and $\Delta = 0$, so that $S_3 \neq 0$ but $S_r = 0$ for all $r \geq 4$. The shadow metric is a perfect square: $Q_L(t) = (2\kappa + 3\alpha t)^2$.

Examples. Affine Kac–Moody $V_k(\mathfrak{g})$ for every simple Lie algebra $\mathfrak{g}$ and every non-critical level $k \neq -h^\vee$; type A ($\mathfrak{sl}_N$), types B, C, D (orthogonal and symplectic), exceptional types E_6, E_7, E_8, G_2, F_4 . For all of these:

$$\kappa(V_k(\mathfrak{g})) = \frac{\dim(\mathfrak{g})(k + h^\vee)}{2h^\vee}.$$

Characterization. The nonzero cubic shadow $S_3 = \alpha$ arises from the Lie bracket (structure constants f_{abc}), while the Jacobi identity forces the quartic shadow $S_4 = 0$ on the primary line. The r -matrix $r(z) = \Omega_{\mathfrak{g}}/((k+h^\vee)z)$ (single simple pole; the bar construction absorbs one power of the OPE pole) encodes all nonlinear structure. The tree-level graph sum terminates at trivalent vertices.

5.3. Class C: Contact ($r_{\max} = 4$).

Definition 5.4. A chirally Koszul algebra $\mathcal{A}$ is *contact* if $\Delta = 0$ on every primary line (so the single-line dichotomy gives $r_{\max}|_L \leq 3$ on each line), but there exists a charged stratum direction with $S_4 \neq 0$ (the quartic contact invariant), and the tower terminates at degree 4: $S_r = 0$ for $r \geq 5$.

Mechanism. Class C escapes the single-line dichotomy (which gives $r_{\max} \in \{2, 3, \infty\}$) via *stratum separation*: the quartic contact invariant Q^{contact} lives on a charged stratum whose self-bracket exits the cyclic deformation complex at degree 5, by rank-one rigidity. On the weight-changing primary line, the tower is abelian ($\alpha = 0, S_4 = 0$); the quartic contact is visible only on the full two-dimensional deformation space.

Examples. The $\beta\gamma$ system at conformal weight λ , with

$$\kappa(\beta\gamma_\lambda) = 6\lambda^2 - 6\lambda + 1 = \frac{1}{2} c(\beta\gamma_\lambda),$$

and its Koszul dual, the bc system at the same weight. At the standard weight $\lambda = 0$ or $\lambda = 1$: $c = 2$ and $\kappa = 1$.

5.4. Class FF: Feigin–Frenkel locus ($\kappa = 0$). The four classes G, L, C, M stratify the non-critical locus $\kappa(\mathcal{A}) \neq 0$. A fifth class appears at the boundary $\kappa = 0$: the *Feigin–Frenkel locus* where the Sugawara topologisation fails.

Definition 5.5 (Class FF). A chirally Koszul algebra $\mathcal{A}$ belongs to class FF if $\kappa(\mathcal{A}) = 0$. The shadow depth $r_{\max}(\mathcal{A})$ is then unconstrained by the single-line dichotomy and is read from the Feigin–Frenkel coset data $\text{coset}_{FF}(\mathcal{A})$.

Examples. The critical-level affine algebra $V_{-h^\vee}(\mathfrak{g})$ for every simple $\mathfrak{g}$ (so $V_{-2}(\mathfrak{sl}_2), V_{-3}(\mathfrak{sl}_3), \dots$); its commutant lies in the Feigin–Frenkel centre $\mathfrak{z}(\widehat{\mathfrak{g}})$, polynomial in infinitely many Sugawara-like generators. The Koszul dual of class L under the involution $k \mapsto -k - 2h^\vee$ passes through class FF at the fixed point $k = -h^\vee$. Critical $\mathcal{W}_N$ (at the Feigin–Frenkel image of the affine critical level) is also class FF.

Characterization. At $\kappa = 0$ the conformal vector degenerates: the Sugawara construction $T = (k + h^\vee)^{-1} : J^a J^a$ has a pole, and topologisation $\text{SC}^{\text{ch}, \text{top}} \rightarrow E_3^{\text{top}}$ fails. The coset slot of the canonical fingerprint (Definition ?? in the monograph) switches from $(\text{Com}(\mathcal{H}_\kappa, \mathcal{A}), \mathcal{A}/\text{Com})$ to the Feigin–Frenkel coset $\text{coset}_{FF}(\mathcal{A}) \subset \mathfrak{z}(\widehat{\mathfrak{g}})$. Class FF is not a shadow-depth class in the sense of classes G, L, C, M : it is the locus on which the ambient shadow invariants degenerate and the classification is controlled by coset data rather than by (α, Δ) .

Remark 5.6 (Why FF is a fifth class, not a refinement). The four classes G, L, C, M are determined by the shadow metric $Q_L(t) = (2\kappa + 3\alpha t)^2 + 2\Delta t^2$, which degenerates at $\kappa = 0$: the leading constant term vanishes, the Riccati algebraicity argument loses its pivot, and the tower of $\Theta_{\mathcal{A}}^{\leq r}$ must be read from a different source. Promoting FF to a fifth companion class is therefore not a refinement of the quadrichotomy but a structural completion: the five classes exhaust the full fingerprint stratum (Theorem ?? in the monograph).

5.5. Class M: Mixed ($r_{\max} = \infty$).

Definition 5.7. A chirally Koszul algebra $\mathcal{A}$ is *mixed* if $\alpha \neq 0$ and $\Delta \neq 0$, so the shadow metric Q_L is an irreducible quadratic in t and the shadow obstruction tower does not terminate.

Examples. Virasoro Vir_c with $\kappa = c/2$; $\mathcal{W}$ -algebras $\mathcal{W}_N$ with $\kappa = (H_N - 1)c$ where $H_N = \sum_{j=1}^N 1/j$; Drinfeld–Sokolov reductions of affine Kac–Moody algebras.

Characterization. The shadow growth rate

$$\rho(\mathcal{A}) = \frac{\sqrt{9\alpha^2 + 16\kappa S_4}}{2|\kappa|}$$

controls the asymptotic behaviour of S_r : the coefficients satisfy $S_r \sim C(\mathcal{A}) \rho(\mathcal{A})^r r^{-5/2} \cos(r\theta + \phi)$ for a branch-point argument θ determined by the shadow metric. The critical central charge c^* where $\rho(\text{Vir}_c) = 1$ is the unique positive real root of the cubic

$$5c^3 + 22c^2 - 180c - 872 = 0, \quad c^* \approx 6.125. \quad (5.1)$$

For $c > c^*$ the tower converges ($\rho < 1$), and for $c < c^*$ it diverges ($\rho > 1$). At the self-dual point $c = 13$: $\rho(\text{Vir}_{13}) \approx 0.467$.

The Virasoro shadow obstruction tower has explicitly computed coefficients through degree 10:

$$\begin{aligned} S_2 &= \frac{c}{2}, & S_3 &= 2, & S_4 &= \frac{10}{c(5c + 22)}, \\ S_5 &= \frac{-48}{c^2(5c + 22)}, & S_6 &= \frac{80(45c + 193)}{3c^3(5c + 22)^2}, & S_7 &= \frac{-2880(15c + 61)}{7c^4(5c + 22)^2}. \end{aligned}$$

The c -independence of $S_3 = 2$ is now proved algebraically by direct residue extraction; see `virasoro_r_matrix.tex`.

Remark 5.8 (Depth and Koszulness). Shadow depth $r_{\max}$ does *not* characterize chiral Koszulness. All four classes (Gaussian, $r_{\max} = 2$; Lie, $r_{\max} = 3$; contact, $r_{\max} = 4$; mixed, $r_{\max} = \infty$) consist of chirally Koszul algebras. Shadow depth classifies *complexity* within the Koszul world: the operadic complexity conjecture identifies $r_{\max}(\mathcal{A})$ with the A_∞ -depth and L_∞ -formality level.

Remark 5.9 (Total depth decomposition). The total shadow depth decomposes as $d(\mathcal{A}) = 1 + d_{\text{arith}} + d_{\text{alg}}$ (Theorem 3.11), where the algebraic depth $d_{\text{alg}} \in \{0, 1, 2, \infty\}$ and depths ≥ 5 are purely arithmetic (cusp forms contribute beyond the algebraic engine).

5.6. The quadrichotomy with logarithmic and Wakimoto companions. The four classes G, L, C, M together with the Feigin–Frenkel locus FF form the fingerprint stratification of the non-logarithmic standard landscape. Two further strata appear once the landscape is enlarged: logarithmic $\mathcal{W}(p)$ and Wakimoto free-field realisations.

Definition 5.10 (Class M^* : logarithmic companion). A chirally Koszul algebra $\mathcal{A}$ belongs to class M^* if it is logarithmic in the sense of Gurarie–Flohr, i.e. the Zhu algebra $A(\mathcal{A})$ is finite-dimensional but the Hochschild Massey products $\langle \Omega, \Omega, \dots, \Omega \rangle$ carry amplitude on arbitrarily high tensor powers of the vacuum.

Examples. Triplet algebras $\mathcal{W}(p)$ for $p \geq 2$; the symplectic-fermion pair at central charge $c = -2$; logarithmic parafermion cosets.

Characterization. Class M^* algebras satisfy $\kappa \neq 0$ and share the mixed-shadow expansion $\alpha \neq 0, \Delta \neq 0$ of class M, but are distinguished by logarithmic amplitudes: Zhu-finite yet Massey-unbounded. The non-logarithmic shadow analysis of Theorem 3.7 applies on the formal-weight line; logarithmic corrections do not affect the algebraic invariants $\alpha = S_3$ and $\Delta = 8\kappa S_4$, which remain finite and computable, but they preclude tempering of the shadow tower.

Definition 5.11 (Class $\mathcal{W}$: Wakimoto stratum). A chirally Koszul algebra $\mathcal{A}$ belongs to class $\mathcal{W}$ if it arises as the screening-kernel of a rank- $(N - 1)$ Heisenberg through Feigin–Frenkel–Wakimoto realisation, $\mathcal{A} = \bigcap_i \ker Q_{\alpha_i} \subset \mathcal{H}^{\oplus(N-1)}$.

Examples. Principal $\mathcal{W}_N$ (as a Wakimoto kernel inside $\mathcal{H}^{\oplus(N-1)}$); affine $\widehat{\mathfrak{sl}}_2$ through the $\beta\gamma\phi$ Wakimoto realisation; admissible affine algebras through parabolic screenings (KRW03, Arakawa07).

Characterization. Class $\mathcal{W}$ is not disjoint from G, L, C, M : it is a presentational overlay. The ambient Heisenberg is class G , and the kernel inherits chiral Koszulity from the screening-kernel preservation theorem. The utility of the class $\mathcal{W}$ label is that it canonicalises the Koszul dual: the Koszul-locus preservation is witnessed explicitly by the screening integrand, and Drinfeld coproducts descend to the intersection from the rank- $(N - 1)$ free-field parent.

Remark 5.12 (Why the full stratification is $G, L, C, M, \text{FF}, M^*, \mathcal{W}$). The four shadow-depth classes G, L, C, M stratify the non-logarithmic non-critical locus by the algebraic invariants (α, Δ) of the shadow metric. Class FF marks the $\kappa = 0$ boundary. Class M^* records logarithmic amplitude behaviour orthogonal to (α, Δ) : a Zhu-finite class-M algebra is class-M, class- M^* , or both, depending on Massey boundedness (Gurarie 1993, Flohr 1996). Class $\mathcal{W}$ records a presentation overlay: a class-L or class-M algebra may simultaneously be class- $\mathcal{W}$ when realised as a screening kernel. The seven-stratum fingerprint is exhaustive on the standard landscape.

5.7. Universal shadow-series exponential base. The asymptotic behaviour of the shadow coefficients on non-logarithmic class M algebras containing a Virasoro subalgebra admits a sharp universal description, independent of the specific family.

Theorem 5.13 (Universal exponential base $C_{\mathcal{A}} = 6$). *Let $\mathcal{A}$ be a non-logarithmic class M chirally Koszul algebra with Virasoro subalgebra at generic central charge. The shadow series $\Sigma_{\mathcal{A}}(z) = \sum_{r \geq 2} S_r(\mathcal{A}) z^r$ has radius of convergence exactly $1/6$; equivalently,*

$$C_{\mathcal{A}} := \limsup_{r \rightarrow \infty} |S_{r+1}(\mathcal{A})/S_r(\mathcal{A})| = 6.$$

The closed form on the Virasoro primary line is

$$\Sigma_{\text{Vir}}(z) = \frac{4z^3}{9} - \frac{z^2}{9} + \frac{z}{27} - \frac{\log(1+6z)}{162}.$$

Pole-doubling for all k : $\Sigma_{\text{Vir}}^{(k)}(z) = [P_k + Q_k \log(1+6z)]/(1+6z)^{2k}$. The constant 6 decomposes as $6 = r_o \cdot S_{r_o} = 3 \cdot 2$: the bar-depth seed times the Riccati seed coefficient.

The constant 6 is programme-universal on class M non-log with Virasoro subalgebra: the same $1/6$ radius appears for $\mathcal{W}_N$ -algebras at leading $1/c$, and the Vol II tempering rate $\beta_{\mathcal{A}}$ coincides with $C_{\mathcal{A}}$ at leading $1/c$. The higher sub-leading $\mathcal{W}_N$ corrections $\beta_N = (N+1)(N+2)/2$ absorb into $C_{\mathcal{W}_N} = 6$ asymptotically.

5.8. Picard–Fuchs spectral-curve presentation. The shadow metric $Q_L(t) = (2\kappa + 3\alpha t)^2 + 2\Delta t^2$ is a rank-one spectral datum; its square root $\sqrt{Q_L}$ is the Seiberg–Witten-type differential of a spectral cover. On non-logarithmic class M, the shadow generating function $\Sigma_{\mathcal{A}}(z)$ satisfies a Picard–Fuchs equation whose monodromy encodes the Borel geometry of the shadow tower.

Proposition 5.14 (Picard–Fuchs for the shadow series). *Let $\mathcal{A}$ be class M with Virasoro subalgebra at generic central charge. The shadow series $\Sigma_{\mathcal{A}}(z)$ satisfies a second-order Picard–Fuchs equation*

$$(1+6z)^2 \Sigma_{\mathcal{A}}''(z) + P_1(z) \Sigma_{\mathcal{A}}'(z) + P_o(z) \Sigma_{\mathcal{A}}(z) = R_{\mathcal{A}}(z),$$

with $P_o, P_1, R_{\mathcal{A}}$ polynomials of degree at most 2 in z determined by (κ, α, Δ) . The regular-singular point $z = -1/6$ is the universal branch point; the monodromy around this point is the reflection sending $z \mapsto -z/(1+6z)$, realising the Borel-plane involution on the shadow tower.

5.9. The depth gap $d_{\text{alg}} \in \{0, 1, 2, \infty\}$.

Proposition 5.15 (Depth gap). *For every chirally Koszul algebra in the standard landscape, the algebraic depth d_{alg} takes values in $\{0, 1, 2, \infty\}$. The value 3 is impossible.*

Witness table. $d_{\text{alg}} = 0$: free fields, bc ghosts (shadow tower trivial). $d_{\text{alg}} = 1$: Heisenberg and lattice (class G). $d_{\text{alg}} = 2$: $\beta\gamma$ on the conformal-weight line (class C) and affine Kac–Moody (class L). $d_{\text{alg}} = \infty$: Virasoro, $\mathcal{W}_N$ principal (class M).

Mechanism. The gap at 3 is structural. If $d_{\text{alg}} = 3$ there is a nonzero degree-4 MC obstruction with no degree-5 follow-up; the Riccati equation $\Delta = 8\kappa S_4$ combined with the shadow Lie Jacobi identity forces either $\Delta = 0$ (which brings $d_{\text{alg}} \leq 2$) or an infinite tower ($d_{\text{alg}} = \infty$). No intermediate value survives.

6. SHADOW TABLES

All entries in the tables below are verified against the compute layer. The modular characteristic $\kappa(\mathcal{A})$ is the degree-2 projection of the universal MC element $\Theta_{\mathcal{A}}$, and the genus- g free energy is $F_g(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g^{\text{FP}}$ for single-generator (uniform-weight) algebras at all genera, and unconditionally at genus 1 for all families. Here $\lambda_g^{\text{FP}} = \frac{2^{2g-1}-1}{2^{2g-1}} \cdot \frac{|B_{2g}|}{(2g)!}$ are the Faber–Pandharipande intersection numbers.

6.1. Master shadow table. Table 1 records, for each standard family, the modular characteristic κ , the shadow class, the shadow depth $r_{\max}$, the cubic coefficient $\alpha = S_3$, the quartic coefficient S_4 , and the critical discriminant $\Delta = 8\kappa S_4$.

TABLE 1. Master shadow table: standard landscape

Algebra $\mathcal{A}$	$\kappa(\mathcal{A})$	Class	$r_{\max}$	S_3	S_4	Δ
<i>Class G (Gaussian)</i>						
Heisenberg $\mathcal{H}_k$	k	G	2	0	0	0
Lattice V_Λ	$\text{rank}(\Lambda)$	G	2	0	0	0
Free fermion ψ	$1/4$	G	2	0	0	0
bc ghosts (weight λ)	$-(6\lambda^2 - 6\lambda + 1)$	G	2	0	0	0
<i>Class L (Lie/tree)</i>						
$\widehat{\mathfrak{sl}}_2$ at level k	$3(k+2)/4$	L	3	$\neq 0$	0	0
$\widehat{\mathfrak{sl}}_N$ at level k	$(N^2-1)(k+N)/(2N)$	L	3	$\neq 0$	0	0
$\widehat{E}_6$ at level k	$13(k+12)/4$	L	3	$\neq 0$	0	0
$\widehat{E}_7$ at level k	$133(k+18)/36$	L	3	$\neq 0$	0	0
$\widehat{E}_8$ at level k	$62(k+30)/15$	L	3	$\neq 0$	0	0
<i>Class C (Contact)</i>						
$\beta\gamma$ (weight λ)	$6\lambda^2 - 6\lambda + 1$	C	4	0	$0^\dagger$	$0^\dagger$
<i>Class M (Mixed)</i>						
Vir_c	$c/2$	M	∞	2	$\frac{10}{c(5c+22)}$	$\frac{40}{5c+22}$
$\mathcal{W}_3$ at c	$5c/6$	M	∞	$\neq 0$	$\neq 0$	$\neq 0$
$\mathcal{W}_4$ at c	$13c/12$	M	∞	$\neq 0$	$\neq 0$	$\neq 0$
$\mathcal{W}_5$ at c	$77c/60$	M	∞	$\neq 0$	$\neq 0$	$\neq 0$
$\mathcal{W}_N$ at c (general)	$(H_N-1)c$	M	∞	$\neq 0$	$\neq 0$	$\neq 0$

[†]On the primary line, $S_4 = 0$ and $\Delta = 0$ for $\beta\gamma$; the quartic contact invariant $Q^{\text{contact}} \neq 0$ lives on a charged stratum (stratum separation, Definition 5.4).

In the table, $H_N = 1 + 1/2 + \dots + 1/N$ denotes the N -th harmonic number. The Virasoro quartic contact $S_4 = Q_{\text{Vir}}^{\text{contact}} = 10/[c(5c+22)]$ is the unique quartic shadow on the T -line, proved by direct computation from the Virasoro OPE. The T -line shadow data ($S_3 = 2, S_4 = 10/[c(5c+22)]$) is *universal*: it coincides for $\mathcal{W}_N$ at all $N \geq 2$, since every $\mathcal{W}_N$ contains the Virasoro subalgebra along the stress-tensor primary line. The additional primary lines (spin-3, spin-4, etc.) carry independent shadow data with separate quartic coefficients.

6.2. Exceptional and non-simply-laced types. Table 2 records the affine Kac–Moody data for exceptional and non-simply-laced types. All are class L.

The kappa formula is universal: $\kappa(V_k(\mathfrak{g})) = \dim(\mathfrak{g}) (k + h^\vee)/(2h^\vee)$. The central charge sum $c + c' = 2 \dim(\mathfrak{g})$ under the Feigin–Frenkel involution $k \mapsto -k - 2h^\vee$, and the complementarity $\kappa + \kappa' = 0$

TABLE 2. Shadow data for exceptional and non-simply-laced types

$\mathfrak{g}$	$\dim(\mathfrak{g})$	$h^\vee$	$\kappa(V_k(\mathfrak{g}))$	$c(V_k) + c(V_{k'})$	Class	$r_{\max}$
$B_2 \cong \mathfrak{so}_3$	10	3	$5(k+3)/3$	20	L	3
$C_2 \cong \mathfrak{sp}_4$	10	3	$5(k+3)/3$	20	L	3
G_2	14	4	$7(k+4)/4$	28	L	3
F_4	52	9	$26(k+9)/9$	104	L	3
E_6	78	12	$13(k+12)/4$	156	L	3
E_7	133	18	$133(k+18)/36$	266	L	3
E_8	248	30	$62(k+30)/15$	496	L	3

holds for all affine Kac–Moody algebras. For non-simply-laced types, the dual Coxeter number $h^\vee$ differs from the Coxeter number h ($h^\vee = 3$ for B_2 and C_2 , while $h = 4$). The shadow class is L regardless: the Jacobi identity kills the quartic on every primary line.

Remark 6.1 (B_2/C_2 isomorphism). The Lie algebras $B_2 = \mathfrak{so}_3$ and $C_2 = \mathfrak{sp}_4$ are isomorphic, sharing $\dim = 10$, $h^\vee = 3$, and $h = 4$. Their Cartan matrices are transposes of each other, but all shadow invariants coincide.

TABLE 3. Complementarity data: $\kappa(\mathcal{A}) + \kappa(\mathcal{A}^\dagger)$ and $c(\mathcal{A}) + c(\mathcal{A}^\dagger)$ under Koszul duality

Family $\mathcal{A}$	Koszul dual $\mathcal{A}^\dagger$	$c + c'$	$\kappa + \kappa'$	Self-dual point
$\mathcal{H}_k$	$\text{Sym}^{\text{ch}}(V^*)$	$\ddagger$	o	none
V_Λ (lattice)	$V_{\Lambda^\dagger}$	$\ddagger$	o	Λ self-dual
Free fermion	$\text{Sym}^{\text{ch}}(\gamma)$	o	o	--
$\beta\gamma_\lambda$	bc_λ	o	o	--
$V_k(\mathfrak{sl}_2)$	$V_{-k-4}(\mathfrak{sl}_2)$	6	o	$k = -2$ (critical)
$V_k(\mathfrak{sl}_N)$	$V_{-k-2N}(\mathfrak{sl}_N)$	$2(N^2-1)$	o	$k = -N$ (critical)
$V_k(\mathfrak{g})$ (general)	$V_{-k-2h^\vee}(\mathfrak{g})$	$2 \dim(\mathfrak{g})$	o	$k = -h^\vee$ (critical)
Vir_c	Vir_{26-c}	26	13	$c = 13$
$\mathcal{W}_3$ at c	$\mathcal{W}_3$ at $100-c$	100	$250/3$	$c = 50$
$\mathcal{W}_4$ at c	$\mathcal{W}_4$ at $246-c$	246	$533/2$	$c = 123$
$\mathcal{W}_5$ at c	$\mathcal{W}_5$ at $488-c$	488	$9394/15$	$c = 244$
$\mathcal{W}_N$ at c (general)	$\mathcal{W}_N$ at K_N-c	K_N	$(H_N-1) K_N$	$c = K_N/2$

6.3. Complementarity table. $\ddagger$ The Koszul dual $\mathcal{H}_k^\dagger = \text{Sym}^{\text{ch}}(V^*)$ is not a Heisenberg algebra: $\kappa(\mathcal{H}_k^\dagger) = -k = \kappa(\mathcal{H}_{-k})$, but the OPE structure of $\mathcal{H}_k^\dagger$ differs from that of $\mathcal{H}_{-k}$. The central charge sum is not defined within a single-parameter family.

Here $K_N = 2(N-1)(2N^2+2N+1)$ is the Koszul conductor for $\mathcal{W}_N = \mathcal{W}(\mathfrak{sl}_N)$ (the sum $c + c'$ under the Feigin–Frenkel involution $k \mapsto -k - 2N$ on the underlying $\mathfrak{sl}_N$). The key structural distinction: for all affine Kac–Moody algebras, $\kappa + \kappa' = 0$ (anti-symmetry under the Feigin–Frenkel involution);

for $\mathcal{W}$ -algebras, $\kappa + \kappa' = (H_N - 1) \cdot K_N \neq 0$. The Virasoro ($N = 2$) gives $\kappa + \kappa' = \frac{1}{2} \cdot 26 = 13$, a nonzero constant independent of c . The Koszul involution $c \mapsto 26 - c$ has a fixed point at $c = 13$, where $\kappa(\text{Vir}_{13}) = 13/2$ and Vir_{13} is self-dual under uncurved quadratic duality.

6.4. Genus tables. The Faber–Pandharipande numbers and the genus expansion $F_g(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g^{\text{FP}}$ (uniform-weight).

TABLE 4. Faber–Pandharipande intersection numbers λ_g^{FP}

g	λ_g^{FP}	Decimal
1	$1/24$	$0.041\overline{6}$
2	$7/5760$	1.215×10^{-3}
3	$31/967680$	3.204×10^{-5}
4	$127/154828800$	8.201×10^{-7}
5	$73/3503554560$	2.084×10^{-8}

The formula is $\lambda_g^{\text{FP}} = \frac{2^{2g-1}-1}{2^{2g-1}} \cdot \frac{|B_{2g}|}{(2g)!}$, where B_{2g} is the $2g$ -th Bernoulli number. The generating function is $\sum_{g \geq 1} \lambda_g^{\text{FP}} x^{2g} = \hat{A}(x) - 1$, where $\hat{A}(x) = (x/2)/\sinh(x/2)$ is the $\hat{A}$ -genus (so $\hat{A}(ix) = (x/2)/\sin(x/2)$). The numbers λ_g^{FP} are strictly positive, strictly decreasing, and decay like $(2\pi)^{-2g}$.

TABLE 5. Genus expansion $F_g(\mathcal{A}) = \kappa \cdot \lambda_g^{\text{FP}}$ (uniform-weight) for selected families

Algebra $\mathcal{A}$	κ	F_1	F_2	F_3
$\mathcal{H}_1$ (Heisenberg, $k=1$)	1	$1/24$	$7/5760$	$31/967680$
$V_1(\mathfrak{sl}_2)$	$9/4$	$3/32$	$7/2560$	$31/430080$
$V_1(\mathfrak{sl}_3)$	$16/3$	$2/9$	$7/1080$	$31/181440$
V_Λ (Leech, rank=24)	24	1	$7/240$	$31/40320$
$\beta\gamma$ (standard, $\lambda=1$)	1	$1/24$	$7/5760$	$31/967680$
Vir_1	$1/2$	$1/48$	$7/11520$	$31/1935360$
Vir_{13}	$13/2$	$13/48$	$91/11520$	$403/1935360$
Vir_{26}	13	$13/24$	$91/5760$	$403/967680$
$\mathcal{W}_3$ at $c=6$	5	$5/24$	$7/1152$	$31/193536$

The entries for $V_1(\mathfrak{sl}_3)$ use $\kappa = (8 \cdot 4)/(2 \cdot 3) = 16/3$. The Leech lattice (rank = 24) gives $F_1 = 1$, reflecting the Jacobi Δ -function origin of the Leech theta series.

6.5. Shadow growth rate table. The shadow growth rate $\rho = \sqrt{(9\alpha^2 + 16\kappa S_4)/(4\kappa^2)}$ determines the convergence of the shadow obstruction tower. For Virasoro, $\rho^2 = (180c + 872)/[c^2(5c + 22)]$. The critical central charge $c^* \approx 6.125$ (the positive real root of (5.1)) separates the convergent regime ($c > c^*$, tower summable) from the divergent regime ($c < c^*$, tower grows exponentially). Class G and class L algebras have $\rho = 0$ (the tower terminates, so there is no exponential growth).

TABLE 6. Shadow growth rate $\rho(\mathcal{A})$ for class M algebras (Virasoro at selected central charges)

c	κ	Q^{contact}	ρ	Regime
1/2 (Ising)	1/4	40/49	≈ 12.53	divergent
1	1/2	10/27	≈ 6.24	divergent
4	2	5/84	≈ 1.54	divergent
$c^* \approx 6.125$	≈ 3.06	≈ 0.031	1	critical
13 (self-dual)	13/2	10/1131	≈ 0.467	convergent
25	25/2	2/735	≈ 0.242	convergent
26 (bosonic string)	13	5/1976	≈ 0.232	convergent

TABLE 7. Virasoro shadow coefficients S_r for $r = 2, \dots, 10$

r	$S_r(\text{Vir}_c)$
2	$c/2$
3	2
4	$10/[c(5c + 22)]$
5	$-48/[c^2(5c + 22)]$
6	$80(45c + 193)/[3c^3(5c + 22)^2]$
7	$-2880(15c + 61)/[7c^4(5c + 22)^2]$
8	$80(2025c^2 + 16470c + 33314)/[c^5(5c + 22)^3]$
9	$-1280(2025c^2 + 15570c + 29554)/[3c^6(5c + 22)^3]$
10	$256(91125c^3 + 1050975c^2 + 3989790c + 4969967)/[c^7(5c + 22)^4]$

6.6. Virasoro shadow obstruction tower table. Each S_r is a rational function of c with poles at $c = 0$ and $c = -22/5$; these are the two values where the quadratic Casimir $c(5c + 22)$ vanishes. The sign pattern $S_r > 0$ for even r and $S_r < 0$ for odd $r \geq 5$ (at $c > 0$) reflects the oscillatory decay $\cos(r\theta + \phi)$ from the complex branch points of the shadow generating function. All entries have been verified by two independent methods: closed-form derivation and the convolution recursion from $H^2 = t^4 Q_L$.

7. STRUCTURAL THEOREMS

A classification is not a theorem. Left as a table, the four shadow classes would be an observation: a pattern in the data, subject to the next counterexample. What transforms a table into a theorem is the proof that no other classes exist, the identification of the algebraic mechanism that enforces the table, and the discovery that the classes carry structure beyond the taxonomy itself.

This section establishes five results that accomplish this transformation. The depth gap (Proposition 7.1) proves that $d_{\text{alg}} \in \{0, 1, 2, \infty\}$ with no intermediate values: the classification is forced by the arithmetic of a single quadratic polynomial. The SC-formality characterisation (Proposition 7.2) shows that class **G** is distinguished by a sharp operadic property. The betagamma contact mechanism (Proposition 7.4) reveals how class **C** escapes the single-line dichotomy. The DS escalation conjecture (Conjecture 7.6) identifies the functor that moves between classes. The join-semilattice theorem (Proposition 7.8) shows that the classes carry a natural algebraic order.

7.1. The depth gap theorem. The shadow tower of a chirally Koszul algebra can terminate or diverge. One might expect every finite depth to be realised: that the tower could stop at degree 5, or 7, or 100, by some suitable choice of OPE data. This does not happen. The tower terminates at degree 3 (class **G**), at degree 4 (class **L**), at degree 5 (class **C**, via stratum separation), or not at all (class **M**). No other depth is accessible.

The reason is that the shadow generating function $H(t) = t^2 \sqrt{Q_L(t)}$ is controlled by a quadratic polynomial $Q_L(t)$, and a quadratic polynomial is either a perfect square or it is not. When Q_L is a perfect square, the square root is linear and the tower terminates; when Q_L is not a perfect square, the binomial series for $(1+u)^{1/2}$ produces infinitely many nonzero coefficients. There is no third possibility. This is the Riccati dichotomy: the depth of the shadow tower is not a free parameter but is read off from the discriminant of a single quadratic.

Two distinct notions of depth appear in this analysis and must not be conflated. The algebraic depth $d_{\text{alg}}(\mathcal{A})$ counts independent shadow coefficients: $d_{\text{alg}} = 0$ if $S_r = 0$ for all $r \geq 3$, $d_{\text{alg}} = n$ if $S_r = 0$ for all $r \geq n+3$, and $d_{\text{alg}} = \infty$ if infinitely many S_r are nonzero. The generating depth $d_{\text{gen}}(\mathcal{A})$ is the degree at which higher operations are recursively determined by lower ones. For the Virasoro algebra, $d_{\text{gen}} = 3$ (the cubic $S_3 = 2$ and quartic $S_4 = 10/[c(5c+22)]$ determine all S_r via the convolution recursion) but $d_{\text{alg}} = \infty$ (every S_r is nonzero). The recursion is finitely generated; the output is not finitely supported.

Proposition 7.1 (Depth gap). *Let $\mathcal{A}$ be a chirally Koszul algebra in the standard landscape with $\kappa(\mathcal{A}) \neq 0$. Then*

$$d_{\text{alg}}(\mathcal{A}) \in \{0, 1, 2, \infty\}.$$

*No finite value $d_{\text{alg}} \geq 3$ is realized. The four values correspond bijectively to the shadow classes **G**, **L**, **C**, **M**.*

Proof. The argument splits into a rank-one analysis on each primary line and a multi-sector extension.

Rank one. On a primary line L with $\kappa \neq 0$, the shadow generating function $H(t) = t^2 \sqrt{Q_L(t)}$ is determined by (κ, α, S_4) (Theorem 3.1), where the shadow metric $Q_L(t) = (2\kappa + 3\alpha t)^2 + 2\Delta t^2$ is the Gaussian decomposition and $\Delta = 8\kappa S_4$ is the critical discriminant.

If $\Delta = 0$ (equivalently $S_4 = 0$), then $Q_L(t) = (2\kappa + 3\alpha t)^2$ is a perfect square, and $H(t) = t^2(2\kappa + 3\alpha t)$ is a polynomial of degree 3. The shadow coefficients $S_r = 0$ for all $r \geq 4$, giving $d_{\text{alg}} \in \{0, 1\}$ according to whether $\alpha = 0$ (class **G**) or $\alpha \neq 0$ (class **L**).

If $\Delta \neq 0$ (equivalently $S_4 \neq 0$), then Q_L is not a perfect square in $k(c)[t]$: the interaction correction $2\Delta t^2$ prevents $\sqrt{Q_L}$ from being polynomial. The Taylor expansion of $\sqrt{Q_L(t)}$ about $t = 0$ has infinitely many nonzero coefficients (the binomial series for $(1+u)^{1/2}$ with $u = (6\alpha t + (9\alpha^2 + 2\Delta)t^2)/(4\kappa^2)$ does not terminate when the coefficient of t^2 in u is nonzero). Therefore $S_r \neq 0$ for infinitely many r , and $d_{\text{alg}} = \infty$ (class **M**).

The gap at $d_{\text{alg}} = 3$ is now forced. Suppose $S_4 \neq 0$. The convolution recursion at degree 5 gives $S_5 = -(6/5)(1/\kappa)\alpha S_4$. If $\alpha \neq 0$: this is nonzero, so $S_5 \neq 0$. If $\alpha = 0$: then $S_5 = 0$, but the degree-6 recursion gives $S_6 \propto S_4^2/\kappa \neq 0$. In either case, $S_4 \neq 0$ propagates to infinitely many nonzero S_r : the closed form $H(t) = t^2 \sqrt{Q_L(t)}$ makes this manifest. Termination at any finite degree ≥ 5 is impossible.

Class **C** ($d_{\text{alg}} = 2$) occupies the boundary where $\Delta = 0$ on the primary line but $S_4 \neq 0$ on a charged stratum: stratum separation (Remark ??) terminates the tower at degree 4 by a mechanism external to the rank-one Riccati analysis. See Proposition 7.4 below.

Multi-sector extension. For a multi-generator algebra with primary lines $L_1, \dots, L_n$, the cubic pump propagates: if $\alpha \neq 0$ on any mixed sector, the adjoint ad_{S_3} generates an infinite chain on every reachable sector, forcing $d_{\text{alg}} = \infty$. If all mixed cubics vanish, the sectors decouple and $d_{\text{alg}} \leq \max_i d_{\text{alg}}(L_i) \leq 2$. No intermediate value $d_{\text{alg}} = 3$ is accessible. $\square$

7.2. SC-formality characterises class **G.** Every algebra in the standard landscape is chirally Koszul: the bar spectral sequence collapses at E_2 , the bar cohomology concentrates in degree 1, and the transferred A_∞ operations on $H^*(\bar{B}^{\text{ch}}(\mathcal{A}))$ satisfy $m_k = 0$ for $k \geq 3$. Koszulness does not distinguish the four shadow classes.

A stronger condition does. Swiss-cheese formality asks that the higher SC-operations on $\mathcal{A}$ *itself* vanish: $m_k^{\text{SC}} = 0$ for all $k \geq 3$. This is a condition on the algebra, not on its bar cohomology; the common notation m_k for both is a source of systematic confusion. The question is: which algebras satisfy this stronger condition?

The answer is a unique-survivor argument. Four shadow classes enter; one constraint eliminates three.

Proposition 7.2 (SC-formality iff class **G**). *Let $\mathcal{A}$ be a chiral algebra in the standard landscape. Then $\mathcal{A}$ is Swiss-cheese formal if and only if $\mathcal{A}$ belongs to class **G**.*

Proof. The forward direction is immediate. If $\mathcal{A}$ is class **G** (Heisenberg, lattice, free fermion, bc ghosts), the current algebra bracket is central: $[a_m, a_n] = km \delta_{m+n,0}$ for Heisenberg, and the analogous statement for every other class **G** family. All nested brackets vanish, so $m_k^{\text{SC}} = 0$ for $k \geq 3$.

The converse is a three-step forced march. Suppose $\mathcal{A}$ is SC-formal. The cubic shadow S_3 is the normalized projection of the trilinear form

$$C(x, y, z) = \kappa(x, [y, z]),$$

where κ is the invariant bilinear form and $[-, -]$ the current algebra bracket. The chain of implications is:

$$\text{SC-formal} \implies m_3^{\text{SC}} = 0 \implies S_3 = 0 \implies C = 0.$$

The first arrow is the definition. The second holds because S_3 is the degree-3 shadow projection of m_3^{SC} . The third requires the key input: κ is nondegenerate on the standard landscape. With κ nondegenerate, $C = 0$ forces $[y, z] = 0$ for all y, z . The current algebra is abelian; an abelian current algebra at level k is Heisenberg; Heisenberg is class **G**.

Classes **L**, **C**, and **M** all have $S_3 \neq 0$ or carry structure (charged strata, nonlinear composites) incompatible with $m_3^{\text{SC}} = 0$. Each is eliminated. Only class **G** survives. $\square$

Remark 7.3 (The two meanings of formality). The distinction between A_∞ -formality of bar cohomology and SC-formality of the algebra itself is the central subtlety of the Koszulness programme. Every chirally Koszul algebra is A_∞ -formal on its bar cohomology; this is a consequence of the E_2 -collapse and holds in all four shadow classes. SC-formality is a strictly stronger condition, operative on the algebra rather than on its cohomological shadow. Proposition 7.2 identifies the precise locus where the two notions coincide: class **G**.

7.3. The betagamma contact mechanism. The single-line Riccati dichotomy produces three outcomes: $d_{\text{alg}} \in \{0, 1, \infty\}$. Class **C** ($d_{\text{alg}} = 2$) is absent from this list. Its existence requires a mechanism that the rank-one analysis cannot see: shadow data that lives on a charged stratum rather than on the primary line, and a termination mechanism that operates by dimensional obstruction rather than by algebraic identity.

The $\beta\gamma$ system is the algebra that exhibits this mechanism. It is the unique standard family where shadow depth exceeds collision depth: the OPE pole order is $p_{\text{max}} = 1$ (a single simple pole $\beta(z)\gamma(w) \sim 1/(z-w)$), the collision depth is $k_{\text{max}} = 0$ (no collision singularity), yet the shadow depth is $r_{\text{max}} = 4$. Three invariants that agree for every other standard family here separate: $p_{\text{max}} \neq k_{\text{max}} \neq r_{\text{max}}$. The mechanism is stratum separation: the quartic shadow lives on a charged stratum invisible to the single-line dichotomy, and the tower terminates there by rank-one rigidity.

Proposition 7.4 (Contact class via stratum separation). *Let $\beta\gamma_\lambda$ denote the $\beta\gamma$ system at conformal weight λ , with central charge $c(\beta\gamma_\lambda) = 2(6\lambda^2 - 6\lambda + 1)$ and $\kappa(\beta\gamma_\lambda) = 6\lambda^2 - 6\lambda + 1$. Then:*

- (i) *On the weight-changing primary line, the OPE is abelian ($\alpha = 0$) and $\Delta = 0$: the single-line dichotomy gives $r_{\max}|_L = 2$ (class **G** on this line).*
- (ii) *The quartic contact invariant $\mathfrak{Q}_{\beta\gamma} = \text{cyc}(m_3)$ is nonzero and lives in the charged sector $\text{Def}_{\text{cyc}}^{(q)}$ with $q \neq 0$.*
- (iii) *The self-bracket $\text{Def}_{\text{cyc}}^{(q)} \otimes \text{Def}_{\text{cyc}}^{(q)} \rightarrow \text{Def}_{\text{cyc}}^{(2q)}$ has $\dim \text{Def}_{\text{cyc}}^{(2q)} = 0$ for rank-one systems: the quartic pump does not activate, and the tower terminates at $r_{\max} = 4$.*
- (iv) *The shadow depth is $r_{\max}(\beta\gamma_\lambda) = 4$ and $d_{\text{alg}}(\beta\gamma_\lambda) = 2$ for all λ .*

Proof. The four items form a logical chain: (i) shows that the primary line sees nothing beyond class **G**; (ii) identifies where the quartic shadow lives; (iii) explains why the tower stops there; (iv) assembles the classification.

Item (i): invisibility on the primary line. The $\beta\gamma$ OPE $\beta(z)\gamma(w) \sim 1/(z-w)$ is a simple pole. The self-OPEs $\beta(z)\beta(w)$ and $\gamma(z)\gamma(w)$ are regular. There is no current algebra bracket: the OPE on the weight-changing line is abelian, $\alpha = 0$, $S_4 = 0$, $\Delta = 0$. The Riccati dichotomy applied to this line gives $d_{\text{alg}} = 0$. If the primary line were the whole story, $\beta\gamma$ would be class **G**.

Item (ii): birth of the quartic on the charged stratum. The primary line is not the whole story. The $\beta\gamma$ system has two generators of different conformal weight: the cyclic deformation complex decomposes by charge, and the charged sector $\text{Def}_{\text{cyc}}^{(q)}$ with $q \neq 0$ is invisible to the restriction $q = 0$ that defines the primary line. The quartic contact invariant $\mathfrak{Q}_{\beta\gamma} = \text{cyc}(m_3)$ is the trilinear Massey product on this charged stratum (Theorem ??). At conformal weight $\lambda = 1/2$ (the symplectic boson), a $\mathbb{Z}/2$ symmetry forces $\mathfrak{Q} = 0$; at all other λ , this symmetry is broken and $\mathfrak{Q} \neq 0$. The quartic shadow is born.

Item (iii): death of the quintic by rank-one rigidity. The quartic shadow, once born, must either propagate (producing $d_{\text{alg}} = \infty$, class **M**) or die (giving $d_{\text{alg}} = 2$, class **C**). The propagation mechanism is the quartic pump: the self-bracket $\text{Def}_{\text{cyc}}^{(q)} \otimes \text{Def}_{\text{cyc}}^{(q)} \rightarrow \text{Def}_{\text{cyc}}^{(2q)}$. For rank-one systems (one generator in each charge sector), the target space has dimension zero: there is no field of charge $2q$ in a rank-one system. The quintic obstruction S_5 vanishes identically on the charged stratum; the pump has no room to operate. The tower terminates.

Item (iv). Items (i)–(iii) give $S_r = 0$ for $r \geq 5$ on every stratum and $S_4 \neq 0$ on the charged stratum: $r_{\max} = 4$, $d_{\text{alg}} = 2$. The result holds for all λ (the rank-one rigidity is independent of conformal weight), confirming class **C**. $\square$

Remark 7.5 (Why contact is rare). The mechanism that produces class **C** requires three simultaneous conditions: an abelian primary line ($\alpha = 0$), a nonzero quartic on a charged stratum ($\mathfrak{Q} \neq 0$), and rank-one rigidity to terminate the tower ($\dim \text{Def}_{\text{cyc}}^{(2q)} = 0$). The third condition fails for rank ≥ 2 : the target of the quartic pump acquires positive dimension, and the tower generically diverges. Among standard families, only the $\beta\gamma$ system and its Koszul dual bc satisfy all three conditions.

7.4. Shadow depth under Drinfeld–Sokolov reduction. The four shadow classes are static invariants of individual algebras. A natural question is how these invariants transform under the functors that connect families. Quantum Drinfeld–Sokolov reduction is the principal such functor: it sends an affine Kac–Moody algebra $V_k(\mathfrak{g})$ to a $\mathcal{W}$ -algebra $\mathcal{W}^k(\mathfrak{g}, f)$, parameterised by a choice of nilpotent $f \in \mathfrak{g}$. At $f = 0$ the reduction is trivial and the output is the input. At $f \neq 0$ the output is a genuinely different algebra: new composite fields appear (the Sugawara stress tensor T , higher-spin currents $\mathcal{W}_i$), and these composites carry OPE data absent from the affine algebra.

The shadow-theoretic consequence is that DS reduction *escalates* shadow class. An affine Kac–Moody algebra is class **L** ($r_{\max} = 3$): the Jacobi identity kills the quartic shadow on every primary line. Every non-trivial DS reduction creates a Virasoro subalgebra whose quartic shadow $S_4 = 10/[c(5c+22)] \neq 0$ on the stress-tensor line, forcing $\Delta \neq 0$ and hence $r_{\max} = \infty$ by the depth gap. The jump is maximal: from $r_{\max} = 3$ (finite tower, class **L**) to $r_{\max} = \infty$ (infinite tower, class **M**), with no intermediate stop.

Conjecture 7.6 (Shadow depth escalation under DS reduction). *Let $\mathfrak{g}$ be a simple Lie algebra, $f \in \mathfrak{g}$ a non-zero nilpotent, and $k \neq -h^\vee$ a non-critical level. Then*

$$r_{\max}(\mathcal{W}^k(\mathfrak{g}, f)) \geq r_{\max}(V_k(\mathfrak{g})) = 3.$$

*The inequality is strict for every non-zero nilpotent: $r_{\max}(\mathcal{W}^k(\mathfrak{g}, f)) = \infty$ (class **M**) whenever $f \neq 0$.*

The conjecture asserts that DS reduction is a one-way door in shadow complexity: once the nilpotent f is turned on, the tower cannot be made to terminate again. The depth gap (Proposition 7.1) reduces the conjecture to a single computation: it suffices to show $S_4 \neq 0$ on the stress-tensor primary line of $\mathcal{W}^k(\mathfrak{g}, f)$. The infinite tower then follows automatically.

Remark 7.7 (Evidence). Three independent lines of evidence support the conjecture.

(a) *Principal reductions.* For $f = f_{\text{prin}}$ and every simple $\mathfrak{g}$, the conjecture is proved: the principal $\mathcal{W}$ -algebra $\mathcal{W}_N = \mathcal{W}^k(\mathfrak{sl}_N, f_{\text{prin}})$ contains the Virasoro algebra as a subalgebra, and the Virasoro quartic $S_4 \neq 0$ on the T -line is a universal computation independent of $\mathfrak{g}$.

(b) *Non-principal reductions in type A.* Systematic computation verifies the conjecture for all nilpotent orbits of $\mathfrak{sl}_N$ ($N = 3, 4, 5$) at multiple non-critical levels k . Every non-trivial orbit (minimal, subregular, hook-type, and all intermediate) produces class **M**. The minimal nilpotent orbit of $\mathfrak{sl}_3$ (the Bershadsky–Polyakov algebra $\mathcal{W}^k(\mathfrak{sl}_3, f_{\min})$) is the simplest non-principal example: it has a single strong generator beyond the stress tensor, and $S_4 \neq 0$ is verified by direct OPE computation.

(c) *Structural mechanism.* Every non-trivial nilpotent f produces, via the BRST cohomology of the associated quantum DS complex, at least one composite strong generator whose self-OPE has pole order ≥ 4 . The pole order of the composite is bounded below by the Sugawara construction: the stress tensor T always appears as a strong generator, and $T(z)T(w)$ has a quartic pole. This quartic pole creates $S_4 \neq 0$ on the stress-tensor primary line, and the depth gap converts this single nonzero quartic into an infinite tower.

7.5. The join-semilattice of shadow classes. The four shadow classes are invariants of individual algebras. The question of how they behave under algebraic operations requires a separate argument. Direct sum is the simplest such operation: given two chiral algebras $\mathcal{A}_1$ and $\mathcal{A}_2$, their direct sum $\mathcal{A}_1 \oplus \mathcal{A}_2$ has fields from both factors, with OPE between distinct factors regular (no singular terms). What is the shadow class of the direct sum?

The answer is the maximum, and the proof is that the shadow tower decomposes sector by sector when the cross-sector OPE is regular. This gives the four shadow classes the structure of a join-semilattice: a partially ordered set in which every pair of elements has a least upper bound, realised concretely by direct sum.

Proposition 7.8 (Shadow classes form a join-semilattice). *The four shadow classes $\{\mathbf{G}, \mathbf{L}, \mathbf{C}, \mathbf{M}\}$ form a join-semilattice under the total order $\mathbf{G} < \mathbf{L} < \mathbf{C} < \mathbf{M}$, where the join is realised by direct sum of chiral algebras:*

$$\text{class}(\mathcal{A}_1 \oplus \mathcal{A}_2) = \max(\text{class}(\mathcal{A}_1), \text{class}(\mathcal{A}_2)).$$

Equivalently, the shadow depth satisfies $r_{\max}(\mathcal{A}_1 \oplus \mathcal{A}_2) = \max(r_{\max}(\mathcal{A}_1), r_{\max}(\mathcal{A}_2))$.

Proof. The OPE of $\mathcal{A}_1 \oplus \mathcal{A}_2$ decomposes into three sectors: pure $\mathcal{A}_1$, pure $\mathcal{A}_2$, and mixed. The mixed sector has regular OPE by the definition of direct sum: fields from distinct factors are mutually local

with no singular terms. Regular OPE contributes no shadow coefficients at any degree. The shadow tower of $\mathcal{A}_1 \oplus \mathcal{A}_2$ is therefore the direct sum of the towers of $\mathcal{A}_1$ and $\mathcal{A}_2$, and the shadow depth is the maximum of the two.

The decisive case is $\mathbf{L} \oplus \mathbf{L}$: does combining two affine Kac–Moody factors produce a quartic shadow? The answer is no. The quartic obstruction S_4 is intrinsic to each factor’s OPE. On the primary line of $V_k(\mathfrak{g}_1)$, the Jacobi identity kills S_4 ; on the primary line of $V_{k'}(\mathfrak{g}_2)$, the same. The mixed channel $\mathfrak{g}_1 \otimes \mathfrak{g}_2$ contributes no singular OPE terms, so no quartic shadow is born in the cross-sector. Therefore $\mathbf{L} \oplus \mathbf{L} = \mathbf{L}$ (not $\mathbf{C}$, not $\mathbf{M}$): the Jacobi identity of each factor survives the direct sum. $\square$

Remark 7.9 (Monotonicity and its failure). The join-semilattice property is specific to direct sums: the functor $\oplus$ preserves the regularity of cross-sector OPE. Constructions that create new singular terms between sectors violate monotonicity and can raise the shadow class strictly. Two examples illustrate the failure.

Drinfeld–Sokolov reduction: $V_k(\mathfrak{g})$ is class $\mathbf{L}$, but $\mathcal{W}^k(\mathfrak{g}, f)$ is class $\mathbf{M}$ for every non-zero nilpotent f (Conjecture 7.6, proved for principal reductions). The BRST construction creates composite fields with new OPE singularities; these singularities produce the quartic shadow that the Jacobi identity had killed.

Coset construction: $\mathcal{A}/\mathcal{A}'$ can land in a higher class than either factor. The Goddard–Kent–Olive coset of $V_k(\mathfrak{sl}_2)$ by its Cartan subalgebra produces the parafermion algebra, whose shadow depth depends on the coset data rather than on the shadow classes of the two inputs.

Shadow depth is monotone under direct sum, anti-monotone under no operation, and unconstrained under quotient or reduction. The join-semilattice records the stable part of this structure; the unstable part is the subject of Conjecture 7.6.

Remark 7.10 (Relationship to companion papers). The W_3 composite field identity $\Lambda_0|h\rangle = h^2 - 9h/5$ and the complete rank-2 holographic datum are developed in the companion paper *The first rank-2 holographic datum: $\mathcal{H}(W_3)$* (w3_holographic_datum.tex).

Remark 7.11 (Wave-14 reconstitution anchor). This note records the graph-by-graph computation that produces the genus-2 cross-channel correction $\delta F_2(W_3) = (c + 204)/(16c)$ and the gravitational-coproduct primitivity for Vir. Two canonical compute modules carry the independent verification paths: w3_genus2.py (six-graph tabulation at seven test central charges) and ds_hpl_primitivity.py (ghost-number obstruction check). The exceptional-Lie dimensions used elsewhere in the monograph are not re-hardcoded here; when an exceptional dimension appears it is imported from compute/lib/bc_exceptional_categorical_zeta_engine.py at FUNDAMENTAL_DIMS['E8']. Cross-reference to Vol I theorem status: thm:multi-weight-genus-expansion gives $\delta F_2(W_3) = (c + 204)/(16c)$ explicit (proved, not conjectural); thm:grav-primitivity-standalone is the ghost-number-obstruction argument repeated here in three steps. The conjectural extension to general principal $\mathcal{W}(\mathfrak{g})$ is Conjecture 9.3.

8. THE GENUS-2 FREE ENERGY OF W_3

The preceding sections treated the single-channel regime: one strong generator, one primary line, one modular characteristic κ . We now compute the first multi-channel invariant. The W_3 algebra has two strong generators (the stress tensor T of weight 2 and the spin-3 current W of weight 3), and $F_2(W_3)$ receives contributions from both channels and their interactions.

8.1. The Frobenius algebra of the bar-complex CohFT. The bar-complex CohFT of W_3 has a two-dimensional Frobenius algebra with basis $\{e_T, e_W\}$ and diagonal metric from the per-channel modular characteristics:

$$\eta_{TT} = \kappa_T = \frac{c}{2}, \quad \eta_{WW} = \kappa_W = \frac{c}{3}, \quad \eta_{TW} = 0.$$

The total modular characteristic is $\kappa(W_3) = \kappa_T + \kappa_W = 5c/6 = c \cdot (H_3 - 1)$, where $H_3 = 1 + 1/2 + 1/3 = 11/6$ is the third harmonic number. The inverse metric gives the propagators:

$$\eta^{TT} = \frac{2}{c}, \quad \eta^{WW} = \frac{3}{c}.$$

Both propagators have weight 1: the bar-complex propagator is $d \log E(z, w)$, which has weight 1 regardless of the conformal weight of the field being sewed.

The OPE determines the three-point structure constants. The $\mathbb{Z}_2$ symmetry $W \mapsto -W$ kills all three-point functions with an odd number of W -insertions:

$$C_{TTT} = C_{TWW} = C_{WWT} = c, \quad C_{TTW} = C_{TWT} = C_{WTT} = C_{WWW} = 0.$$

The derivations are: $T_{(1)}T = 2T$ gives $C_{TT}^T = 2$, whence $C_{TTT} = \eta_{TT} \cdot 2 = c$; $T_{(1)}W = 3W$ gives $C_{TW}^W = 3$, whence $C_{TWW} = \eta_{WW} \cdot 3 = c$; $W_{(3)}W = 2T$ gives $C_{WW}^T = 2$, whence $C_{WWT} = \eta_{TT} \cdot 2 = c$.

The genus-0 four-point vertex is universal: for any channel pair $(i, j) \in \{T, W\}^2$,

$$V_{0,4}(i, i \mid j, j) = \sum_m \eta^{mm} C_{iim} C_{jjm} = \eta^{TT} \cdot c \cdot c = 2c. \quad (8.1)$$

Only the T -channel contributes to the sum: $C_{iiT} = c$ for $i \in \{T, W\}$, while $C_{iiW} = 0$ for both $i = T$ (since $C_{TTW} = 0$) and $i = W$ (since $C_{WWW} = 0$). The W -channel is invisible at genus-0 valence 4.

8.2. The seven stable graphs at genus 2. The moduli space $\overline{\mathcal{M}}_{2,0}$ has a stratification by seven stable graphs $\Gamma_0, \dots, \Gamma_6$:

	Name	Vertices	Edges	Aut	h^1	Type
Γ_0	smooth	$(g=2, \text{val}=0)$	0	1	0	compact stratum
Γ_1	figure-eight	$(g=1, \text{val}=2)$	1 self-loop	2	1	one-loop
Γ_2	banana	$(g=0, \text{val}=4)$	2 self-loops	8	2	two-loop sunset
Γ_3	dumbbell	$(g=1, \text{val}=1) + (g=1, \text{val}=1)$	1 bridge	2	0	separating
Γ_4	theta	$(g=0, \text{val}=3) + (g=0, \text{val}=3)$	3 bridges	12	2	non-separating
Γ_5	tadpole	$(g=0, \text{val}=3) + (g=1, \text{val}=1)$	1 self-loop + 1 bridge	2	1	mixed
Γ_6	chain	$(g=0, \text{val}=3) + (g=0, \text{val}=3)$	2 self-loops + 1 bridge	8	2	non-separating

Every vertex satisfies the stability condition $2g_v + \text{val}_v \geq 3$, and the genus formula $h^1(\Gamma) + \sum_v g_v = 2$ holds in each case. Note that the smooth graph Γ_0 has no edges and contributes the integrated class on $\mathcal{M}_{2,0}$ itself; the remaining six graphs are the boundary strata.

8.3. Multi-channel Feynman rules. For each graph Γ with $|E(\Gamma)|$ edges, a *channel assignment* is a function $\sigma: E(\Gamma) \rightarrow \{T, W\}$. The Feynman rules of the bar-complex CohFT assign to each pair (Γ, σ) an amplitude

$$\mathcal{A}(\Gamma, \sigma) = \frac{1}{|\text{Aut}(\Gamma)|} \prod_{e \in E(\Gamma)} \eta^{\sigma(e), \sigma(e)} \prod_{v \in V(\Gamma)} V_v(g_v; \sigma|_{\text{half-edges}(v)}),$$

where the vertex factors are:

- *Genus-0, trivalent*: $V_v = C_{\sigma(e_1)\sigma(e_2)\sigma(e_3)}$.
- *Genus-0, four-valent* (the banana vertex): $V_v = \sum_m \eta^{mm} C_{\sigma(e_1)\sigma(e_2)m} C_{m\sigma(e_3)\sigma(e_4)}$.
- *Genus-1, valence n* : $V_v = \kappa_{\sigma(e)}/24$ (per-channel genus-1 universality, proved unconditionally).

Each half-edge at a self-loop contributes a pair of half-edges to the same vertex, both carrying channel $\sigma(e)$.

8.4. Per-graph amplitudes. We compute each boundary graph amplitude, decomposed into *diagonal* (all edges in one channel) and *mixed* (edges in different channels) contributions.

Γ_1 : *figure-eight*. One edge, two channel assignments. The vertex is genus-1 with valence 2, so the vertex factor for channel i is $\kappa_i/24$:

$$\begin{aligned}\mathcal{A}(\Gamma_1, T) &= \frac{1}{2} \eta^{TT} \cdot \frac{\kappa_T}{24} = \frac{1}{2} \cdot \frac{2}{c} \cdot \frac{c}{48} = \frac{1}{48}, \\ \mathcal{A}(\Gamma_1, W) &= \frac{1}{2} \eta^{WW} \cdot \frac{\kappa_W}{24} = \frac{1}{2} \cdot \frac{3}{c} \cdot \frac{c}{72} = \frac{1}{48}.\end{aligned}$$

Both channels contribute equally: the propagator $1/\kappa_i$ cancels the genus-1 vertex factor $\kappa_i/24$, leaving the universal value $1/48$ independent of c . The total is $1/24 = \lambda_1^{\text{FP}}$. No mixed channel assignments exist (single edge).

Γ_2 : *banana*. Two edges, four channel assignments, $|\text{Aut}| = 8$. The vertex is genus-0 with valence 4, and the four-point vertex factor is $V_{0,4}(i, i \mid j, j) = 2c$ universally (8.1).

$$\begin{aligned}\mathcal{A}(\Gamma_2, TT) &= \frac{1}{8} (\eta^{TT})^2 \cdot 2c = \frac{1}{8} \cdot \frac{4}{c^2} \cdot 2c = \frac{1}{c}, \\ \mathcal{A}(\Gamma_2, WW) &= \frac{1}{8} (\eta^{WW})^2 \cdot 2c = \frac{1}{8} \cdot \frac{9}{c^2} \cdot 2c = \frac{9}{4c}, \\ \mathcal{A}(\Gamma_2, TW+WT) &= \frac{1}{8} \cdot 2 \cdot \eta^{TT} \eta^{WW} \cdot 2c = \frac{1}{8} \cdot 2 \cdot \frac{6}{c^2} \cdot 2c = \frac{3}{c}.\end{aligned}$$

The cross-channel contribution is $\delta_{\text{banana}} = 3/c$.

Γ_3 : *dumbbell*. One bridge edge, two channel assignments, $|\text{Aut}| = 2$. Each vertex has genus 1 and valence 1, with vertex factor $\kappa_i/24$:

$$\begin{aligned}\mathcal{A}(\Gamma_3, T) &= \frac{1}{2} \eta^{TT} \cdot \left(\frac{\kappa_T}{24}\right)^2 = \frac{1}{2} \cdot \frac{2}{c} \cdot \frac{c^2}{2304} = \frac{c}{2304}, \\ \mathcal{A}(\Gamma_3, W) &= \frac{1}{2} \eta^{WW} \cdot \left(\frac{\kappa_W}{24}\right)^2 = \frac{1}{2} \cdot \frac{3}{c} \cdot \frac{c^2}{5184} = \frac{c}{3456}.\end{aligned}$$

The total is $c/2304 + c/3456 = (3c + 2c)/6912 = 5c/6912 = \kappa(W_3)/1152$. No mixed channel assignments exist (single edge).

Γ_4 : *theta*. Three bridge edges, eight channel assignments, $|\text{Aut}| = 12$. Each vertex has genus 0 and valence 3, with vertex factor $C_{\sigma(e_1)\sigma(e_2)\sigma(e_3)}$.

- *All-T*: both vertices get $C_{TTT} = c$. Amplitude $= \frac{1}{12} (\eta^{TT})^3 \cdot c^2 = \frac{1}{12} \cdot \frac{8}{c^3} \cdot c^2 = \frac{2}{3c}$.
- *All-W*: both vertices get $C_{WWW} = 0$. Amplitude $= 0$.
- *Type (T, T, W)* (three such assignments): each vertex receives one W half-edge, giving $C_{TTW} = 0$. All three vanish.
- *Type (T, W, W)* (three such assignments): each vertex receives half-edges (T, W, W) , giving $C_{TWW} = c$. The propagator product is $\eta^{TT}(\eta^{WW})^2 = (2/c)(9/c^2) = 18/c^3$. Each assignment contributes $18c^2/c^3 = 18/c$; with $|\text{Aut}| = 12$, the total mixed amplitude is

$$\delta_{\text{theta}} = \frac{3 \cdot 18}{12c} = \frac{9}{2c}.$$

Γ_5 : *tadpole*. One self-loop at vertex v_0 (genus 0, valence 3) and one bridge to vertex v_1 (genus 1, valence 1). Two edges, four channel assignments, $|\text{Aut}| = 2$.

- (T, T) : self-loop T , bridge T . Vertex v_0 gets half-edges $(T, T, T) \rightarrow C_{TTT} = c$. Vertex v_1 gets half-edge $T \rightarrow \kappa_T/24 = c/48$. Propagators: $(\eta^{TT})^2 = 4/c^2$. With $|\text{Aut}|: \frac{1}{2} \cdot \frac{4}{c^2} \cdot c \cdot \frac{c}{48} = \frac{1}{24}$.
- (W, W) : self-loop W , bridge W . Vertex v_0 gets $(W, W, W) \rightarrow C_{WWW} = 0$. Vanishes.
- (T, W) : self-loop T , bridge W . Vertex v_0 gets $(T, T, W) \rightarrow C_{TTW} = 0$. Vanishes.
- (W, T) : self-loop W , bridge T . Vertex v_0 gets $(W, W, T) \rightarrow C_{WWT} = c$. Vertex v_1 gets $T \rightarrow \kappa_T/24 = c/48$. Propagators: $\eta^{WW} \cdot \eta^{TT} = (3/c)(2/c) = 6/c^2$. With $|\text{Aut}|$:

$$\delta_{\text{tadpole}} = \frac{1}{2} \cdot \frac{6}{c^2} \cdot c \cdot \frac{c}{48} = \frac{6c^2}{96c^2} = \frac{1}{16}.$$

The tadpole mixed amplitude $1/16$ is *independent* of c : the c^2 in the numerator (from $C_{WWT} \cdot \kappa_T/24$) cancels the c^2 in the denominator (from $\eta^{WW} \cdot \eta^{TT}$).

Γ_6 : *chain (barbell)*. Two genus-0 trivalent vertices, each carrying one self-loop, connected by a bridge. Three edges (self-loop s_1 , self-loop s_2 , bridge b), eight channel assignments, $|\text{Aut}| = 8$. Each vertex v_j receives half-edges $(\sigma(s_j), \sigma(s_j), \sigma(b))$ with vertex factor $C_{\sigma(s_j) \sigma(s_j) \sigma(b)}$.

Only $\sigma(b) = T$ gives nonzero vertex factors (since $C_{TTW} = C_{WWW} = 0$). Among the four remaining assignments $(\sigma(s_1), \sigma(s_2)) \in \{T, W\}^2$ with $\sigma(b) = T$:

- (T, T, T) : diagonal. Vertex factors $C_{TTT}^2 = c^2$, propagators $(\eta^{TT})^3 = 8/c^3$. With $|\text{Aut}|: (1/8)(8/c^3)(c^2) = 1/c$.
- (T, W, T) : mixed. $C_{TTT} \cdot C_{WWT} = c^2$, $\eta^{TT} \eta^{WW} \eta^{TT} = 12/c^3$. With $|\text{Aut}|: (1/8)(12/c^3)(c^2) = 3/(2c)$.
- (W, T, T) : mixed. By the $s_1 \leftrightarrow s_2$ symmetry: $3/(2c)$.
- (W, W, T) : mixed. $C_{WWT}^2 = c^2$, $(\eta^{WW})^2 \eta^{TT} = 18/c^3$. With $|\text{Aut}|: (1/8)(18/c^3)(c^2) = 9/(4c)$.

The total mixed amplitude is

$$\delta_{\text{barbell}} = \frac{3}{2c} + \frac{3}{2c} + \frac{9}{4c} = \frac{6+6+9}{4c} = \frac{21}{4c}.$$

8.5. The cross-channel correction. Collecting the mixed-channel amplitudes from all boundary graphs:

$$\delta F_2 = \underbrace{\frac{3}{c}}_{\text{banana}} + \underbrace{\frac{9}{2c}}_{\text{theta}} + \underbrace{\frac{1}{16}}_{\text{tadpole}} + \underbrace{\frac{21}{4c}}_{\text{barbell}} = \frac{51}{4c} + \frac{1}{16} = \frac{c+204}{16c}.$$

The figure-eight and dumbbell graphs contribute zero mixed amplitude (single edges admit only diagonal assignments). The correction decomposes into two structurally distinct pieces:

- A c -dependent piece $51/(4c)$, arising from genus-0 vertex factors on the banana ($3/c$), theta ($9/(2c)$), and barbell ($21/(4c)$) graphs. This is R -matrix independent: genus-0 vertices receive no corrections from the CohFT R -matrix.
- A c -independent piece $1/16$, arising from the tadpole graph, which mixes a genus-0 vertex factor with a genus-1 vertex factor $\kappa_i/24$.

Remark 8.1 (Comparison with the scalar approximation). The per-channel (diagonal) graph sum, computed by applying Theorem D to each channel independently, gives

$$F_2^{\text{scal}}(W_3) = \kappa \cdot \lambda_2^{\text{FP}} = \frac{5c}{6} \cdot \frac{7}{5760} = \frac{7c}{6912}.$$

The ratio of the naive cross-channel correction to the scalar approximation is

$$\frac{\delta F_2}{F_2^{\text{scal}}} = \frac{432(c + 204)}{7c^2},$$

which diverges as $c \rightarrow 0$ and decays as c^{-1} at large c . At $c = 4$: the ratio is approximately 802, and the cross-channel correction *dominates* the scalar term. At $c = 50$: the ratio is approximately 6.3. At large c , the tadpole's c -independent contribution $1/16$ dominates, and the correction becomes a subleading $O(1/c)$ effect relative to the $O(c)$ scalar term.

Remark 8.2 (Status of the cross-channel correction). The computation above uses the naive genus-1 vertex factors $\kappa_i/24$ without R -matrix corrections. The multi-generator universality problem (**op:multi-generator-universality**) is resolved negatively: the CohFT R -matrix does *not* produce compensating terms that cancel δF_2 . The cross-channel correction $\delta F_2 = (c+204)/(16c)$ is a genuine additional term, and the correct decomposition is $F_g = \kappa \cdot \lambda_g^{\text{FP}} + \delta F_g^{\text{cross}}$.

8.6. The complete amplitude table. We record the full per-graph data:

Graph	Diagonal (T)	Diagonal (W)	Mixed	Total
Γ_1 figure-eight	$1/48$	$1/48$	0	$1/24$
Γ_2 banana	$1/c$	$9/(4c)$	$3/c$	$25/(4c)$
Γ_3 dumbbell	$c/2304$	$c/3456$	0	$5c/6912$
Γ_4 theta	$2/(3c)$	0	$9/(2c)$	$31/(6c)$
Γ_5 tadpole	$1/24$	0	$1/16$	$5/48$
Γ_6 barbell	$1/c$	0	$21/(4c)$	$25/(4c)$

The table records the six boundary-graph amplitudes; the smooth graph Γ_0 has no edges and contributes an integrated class on $\mathcal{M}_{2,0}$ itself. The mixed column sums to $\delta F_2 = (c + 204)/(16c)$.

Every entry in this table has been verified against the compute module `w3_genus2.py`, using exact rational arithmetic (`fractions.Fraction`), at the seven test central charges $c \in \{1, 2, 4, 13, 26, 50, 100\}$.

8.7. Planted-forest corrections. The planted-forest correction on each primary line is $\delta_{\text{pf}}^{(2,0)} = S_3(10S_3 - \kappa)/48$.

T-line: $S_3^T = 2$ (the Virasoro cubic shadow), $\kappa_T = c/2$.

$$\delta_{\text{pf}}^T = \frac{2(20 - c/2)}{48} = \frac{40 - c}{48}.$$

This vanishes at $c = 40$, is positive for $c < 40$, and negative for $c > 40$.

W-line: $S_3^W = 0$ by the $\mathbb{Z}_2$ parity $W \mapsto -W$, which forces all odd-degree shadows on the W -line to vanish. Hence $\delta_{\text{pf}}^W = 0$.

The total planted-forest correction is $\delta_{\text{pf}} = (40 - c)/48$, living entirely on the T -line.

8.8. Koszul duality constraints. Under the Koszul involution, W_3 at central charge c is dual to W_3 at central charge $c' = 100 - c$ (the Feigin–Frenkel involution for $\mathfrak{sl}_3$). The modular characteristics satisfy

$$\kappa(c) + \kappa(100 - c) = \frac{5c}{6} + \frac{5(100 - c)}{6} = \frac{250}{3},$$

and the scalar free energies satisfy $F_2^{\text{scal}}(c) + F_2^{\text{scal}}(100 - c) = (250/3) \cdot \lambda_2^{\text{FP}}$. The cross-channel correction at the dual pair is

$$\delta F_2(c) + \delta F_2(100 - c) = \frac{c + 204}{16c} + \frac{304 - c}{16(100 - c)} = -\frac{c^2 - 100c - 10200}{8c(100 - c)}.$$

This sum is not constant in c : the cross-channel correction does not respect Koszul additivity. If universality holds (δF_2 is cancelled by R -matrix corrections), this is automatic. If the naive correction persists, it produces a Koszul-asymmetric contribution to F_2 : an observable departure from single-channel behaviour.

9. THE GRAVITATIONAL COPRODUCT IS PRIMITIVE

In gauge theory, gluons split: one gluon can exit a scattering region as two, and the dg-shifted Yangian coproduct encodes this splitting. In gravity, the graviton is composite, not fundamental. The stress tensor is the Sugawara quadratic

$$T = \frac{1}{k+2} \left(\frac{1}{2} : J^0 J^0 : + : J^+ J^- : \right),$$

built from affine currents by normal ordering. This compositeness has a precise algebraic consequence: the transferred coproduct is trivial.

9.1. The DS-HPL transfer theorem. The Drinfeld–Sokolov reduction $\hat{\mathfrak{sl}}_2 \rightarrow \text{Vir}$ is a BRST reduction with a strong deformation retract (i, p, h) from the BRST complex to the Virasoro cohomology. The homotopy h has ghost-number degree -1 : it maps $V \mapsto b$ (ghost number $0 \rightarrow -1$) and $c \mapsto U/2$ (ghost number $1 \rightarrow 0$).

Theorem 9.1 (DS-HPL transfer). *Let $Y^{\text{dg}}(\hat{\mathfrak{sl}}_2)$ denote the dg-shifted Yangian at level k , with A_∞ products $\{m_n^{\text{aff}}\}$, coproduct Δ_z^{aff} , and r -matrix $r^{\text{aff}}(z) = \Omega/((k+2)z)$. The HPL transfer through the BRST SDR produces:*

- (i) *Transferred A_∞ products $\{m_n^{\text{Vir}}\}_{n \geq 2}$ on the Virasoro sector, with $m_n^{\text{Vir}} \neq 0$ for all $n \geq 2$ (the entire infinite A_∞ tower of class **M**).*
- (ii) *A transferred coproduct $\Delta_z^{\text{Vir}} : H_{\text{Vir}} \rightarrow H_{\text{Vir}} \otimes H_{\text{Vir}}$.*
- (iii) *A transferred r -matrix $r^{\text{Vir}}(z) = (c/2)/z^3 + 2T/z$, with cubic leading pole reflecting the Sugawara nonlinearity.*

The affine r -matrix $r^{\text{aff}}(z) = \Omega/((k+2)z)$ has a simple pole (the OPE has at most a double pole; the bar kernel $d\log(z-w)$ absorbs one power). The Virasoro OPE has a quartic pole (z^{-4} in $T(z)T(w)$); absorption produces the cubic pole z^{-3} of $r^{\text{Vir}}(z)$. The general principle: r -matrix pole order = OPE pole order minus one.

Theorem 9.2 (Gravitational coproduct primitivity). *The HPL-transferred coproduct is strictly primitive at all degrees:*

$$\Delta_{z,k}^{\text{Vir}} = 0 \quad \text{for all } k \geq 2.$$

That is, $\Delta_z^{\text{Vir}}(T) = \tau_z(T) \otimes 1 + 1 \otimes T$. All nontrivial structure of the Virasoro dg-shifted Yangian resides in the $r(z)$ -twisted target product, never in the coproduct.

Proof. The proof is a ghost-number obstruction, proceeding in three steps.

Step 1. The linear transferred coproduct $(p \otimes p)(\Delta_z(T))$ is already primitive. The affine coproduct $\Delta_z(T)$ expands as $\tau_z(T) \otimes 1 + 1 \otimes T + \frac{1}{k+2} \Omega(w+z, w)$, where $\Omega(w+z, w) = J^0(w+z) \otimes J^0(w) + J^+(w+z) \otimes J^-(w) + J^-(w+z) \otimes J^+(w)$. The projection p annihilates J^0 (not in the Virasoro sector) and J^+ decomposes as $V + 1$ with $p(V) = 0$. Every summand of Ω contains at least one factor from

$\{J^0, J^-, J^+\}$ at conformal weight 1, where the Virasoro sector has no states ($L_{-1}|0\rangle = 0$). Each tensor factor of $(p \otimes p)(\Omega)$ has p killing at least one component, so $(p \otimes p)(\Omega) = 0$.

Step 2. At degree $k \geq 2$, every HPL tree has at least one internal edge labelled by the homotopy h . The ghost-number accounting is:

Operation	Ghost degree
h (homotopy)	-1
δ (BRST perturbation)	$+1$
μ (vertex algebra product)	0
Δ_z (coproduct)	0

The product $h\delta$ has ghost degree $-1 + 1 = 0$, but the tree topology at degree $k \geq 2$ always contains one *uncompensated* δ : a binary tree with k leaves has $k - 1$ internal vertices (each carrying δ , ghost $+1$) and $k - 2$ internal edges (each carrying h , ghost -1), giving total ghost number $(k - 1) - (k - 2) = +1$. The overall output before projection has ghost number $+1$.

Step 3. The W-algebra cohomology sits in ghost number 0. The projection p annihilates all ghost-number- $(\neq 0)$ elements. Since the output of each degree- k tree has ghost number $+1$, the projection $(p \otimes p)$ kills every summand.

The perturbation does not produce corrections. The BRST perturbation δ has ghost degree $+1$, so the product $h\delta$ preserves ghost number. But $\delta(T) = 0$ (the stress tensor is BRST-closed), so $(h\delta)(i(T)) = 0$, and by induction $(h\delta)^n(i(T)) = 0$ for all $n \geq 1$. The corrected inclusion $i_\infty(T) = i(T)$ is unperturbed, and the same primitive coproduct results at every perturbative order. $\square$

Conjecture 9.3 (General DS primitivity). *For any simple Lie algebra $\mathfrak{g}$, the HPL-transferred coproduct of the principal $\mathcal{W}$ -algebra $\mathcal{W}(\mathfrak{g})$ is strictly primitive at all degrees. The ghost-number obstruction of the BRST complex for the principal Drinfeld–Sokolov reduction $V_k(\mathfrak{g}) \rightarrow \mathcal{W}_k(\mathfrak{g})$ forces $\Delta_{z,n}^{\mathcal{W}} = 0$ for $n \geq 2$.*

9.2. Physical meaning. The coproduct Δ_z of a dg-shifted Yangian encodes the *splitting* of excitations. Its strictness or nontriviality is the algebraic fingerprint of the distinction between fundamental and composite fields.

Theory	Boundary algebra	Δ_z	Shadow depth	Reason
Chern–Simons	$\hat{\mathfrak{g}}_k$ (affine KM)	$\neq \varepsilon$	3 (class L)	fundamental gluons split
3d gravity	$\text{Vir}_c = \mathcal{W}_k(\mathfrak{sl}_2)$	$= \varepsilon$	∞ (class M)	ghost obstruction (DS)
Twisted holography	H_k (Heisenberg)	$= \varepsilon$	2 (class G)	free field, $r = k/z$
M2 brane	DDCA	$\neq \varepsilon$	--	no DS, fundamental modes
M5 brane	$W_{1+\infty}(K)$	$= \varepsilon$	∞ (class M)	ghost obstruction (DS)

The pattern: *principal Drinfeld–Sokolov reductions produce primitive coproducts*. The BRST ghost sector provides the grading obstruction that kills all higher-degree components. Theorem 9.2 proves this for $\widehat{\mathfrak{sl}}_2 \rightarrow \text{Vir}$; the same argument applies to any principal DS reduction $\widehat{\mathfrak{g}} \rightarrow \mathcal{W}(\mathfrak{g})$ (the ghost-number grading is universal), yielding Conjecture 9.3. Theories with fundamental excitations (Chern–Simons, deformed double current algebras) lack a ghost sector, and their coproducts are nontrivial.

The two exceptions to the DS pattern are illuminating. The Heisenberg algebra has a primitive coproduct for a different reason: it is a free field, its r -matrix $r^H(z) = k/z$ is scalar, and the bar complex is formal at degree 2 (shadow depth $r_{\max} = 2$, class **G**). No DS reduction and no ghost obstruction; primitivity follows from the absence of nonlinear OPE structure. The M2 brane boundary algebra (a

deformed double current algebra) has a nontrivial coproduct because it is not obtained by DS reduction from any affine algebra; its excitations are fundamental modes of the M2 worldvolume theory.

9.3. The r -matrix carries all gravitational structure. With the coproduct primitive, the full gravitational content of the Virasoro dg-shifted Yangian is encoded in the r -matrix

$$r^{\text{Vir}}(z) = \frac{c/2}{z^3} + \frac{2T}{z}.$$

The two terms encode distinct physical data:

- The *leading coefficient* $c/2 = \kappa(\text{Vir}_c)$ is the modular characteristic. It determines the genus tower $F_g = (c/2) \int_{\overline{\mathcal{M}}_g} \lambda_g$ (uniform-weight) at all genera, the BTZ black hole entropy $S_{\text{BTZ}} = 2\pi\sqrt{ch/6}$, and the Hawking–Page transition at $h = c/24$.
- The *subleading term* $2T/z$ encodes the state dependence of gravitational scattering. The braiding of Wilson lines (gravitational line operators) in the holomorphic-topological bulk is controlled by this term.
- The *pole structure* (cubic leading pole, simple subleading pole, no even poles) is the signature of a bosonic algebra extracted through the $d\log$ kernel: the OPE poles z^{-4}, z^{-2}, z^{-1} each lose one order under the $d\log$ extraction, producing r -matrix terms z^{-3}, z^{-1}, z^0 . The z^0 term (from the ∂T pole at z^{-1}) is regular and does not contribute to the r -matrix, leaving only z^{-3} and z^{-1} .

The contrast with Chern–Simons is maximal. In Chern–Simons gauge theory, the affine r -matrix $r^{\text{aff}}(z) = \Omega/((k+h^\vee)z)$ is a simple pole and the coproduct is nontrivial: gluons split and the splitting is nontrivial. In gravity, the r -matrix has a cubic pole and the coproduct is trivial: gravitons braid but do not split. The Drinfeld–Sokolov reduction bridges the two: it takes the affine Casimir $\Omega/((k+h^\vee)z)$ to the gravitational $(c/2)/z^3 + 2T/z$ by homological perturbation through the Sugawara nonlinearity, and in the same stroke, the ghost-number obstruction kills the nontrivial coproduct.

Datum	Chern–Simons	Gravity
boundary algebra	$V_k(\mathfrak{g})$ (affine KM)	Vir_c (Virasoro)
highest OPE pole	z^{-2} (quadratic)	z^{-4} (quartic)
$\mathcal{A}_\infty$ tower	strict ($m_{k \geq 3} = 0$)	infinite ($m_k \neq 0$ all k)
shadow depth	3 (class L)	∞ (class M)
$r^{\text{coll}}(z)$	$\Omega/((k+h^\vee)z)$ (simple pole)	$(c/2)/z^3 + 2T/z$ (cubic pole)
Δ_z	nontrivial (gluons split)	primitive (gravitons braid)
κ	$\dim \mathfrak{g} (k + h^\vee)/(2h^\vee)$	$c/2$
$\kappa + \kappa^\dagger$	0	13

The entire table descends from a single datum: the number of λ -powers in the λ -bracket. Affine Kac–Moody has λ^1 at most (quadratic pole), yielding strict associativity, finite shadow depth, a simple-pole r -matrix, and a nontrivial coproduct. The Virasoro algebra has λ^3 (quartic pole), yielding the full infinite $\mathcal{A}_\infty$ tower, infinite shadow depth, a cubic-pole r -matrix, and a primitive coproduct. The Drinfeld–Sokolov reduction is the functor that connects the two columns: it increases the λ -bracket order by two, trades the nontrivial coproduct for a ghost obstruction, and replaces the complementarity $\kappa + \kappa^\dagger = 0$ with $\kappa + \kappa^\dagger = 13$.

In one sentence: *gravity does not produce particles, only braids; the graviton is composite, and the ghost sector of its composite origin is the algebraic obstruction that kills the splitting.*

10. OPEN PROBLEMS AND OUTLOOK

We collect several directions suggested by the results of this paper.

10.1. Multi-generator universality at genus ≥ 2 . The genus-2 computation for $\mathcal{W}_3$ (Section 8) produces a cross-channel correction $\delta F_2 = (c + 204)/(16c)$ to the scalar approximation $F_2^{\text{scal}} = \kappa \cdot \lambda_2^{\text{FP}}$. This correction is R -matrix independent and is not cancelled by CohFT R -matrix terms: the multi-generator universality problem is resolved negatively at genus 2. The correct decomposition is $F_g = \kappa \cdot \lambda_g^{\text{FP}} + \delta F_g^{\text{cross}}$. At genus 1, the identity $F_1 = \kappa \cdot \lambda_1^{\text{FP}}$ holds unconditionally for all families. The structure of $\delta F_g^{\text{cross}}$ at genus $g \geq 3$ remains to be computed.

10.2. The operadic complexity conjecture. The four-class partition **G/L/C/M** classifies shadow depth. The operadic complexity theorem identifies $r_{\max}(\mathcal{A})$ with the $\mathcal{A}_\infty$ -depth (the minimal n such that $m_k = 0$ for $k > n$ in the transferred $\mathcal{A}_\infty$ structure) and with the L_∞ -formality level. The shadow-formality identification is proved at all degrees by induction on r (thm:shadow-formality-identification).

10.3. Pixton's ideal and the shadow obstruction tower. For class **M** algebras (infinite shadow obstruction tower), the Maurer–Cartan equation at each genus g produces a tautological relation on $\overline{\mathcal{M}}_g$. These relations lie in the Pixton ideal (the ideal of relations among tautological classes), and it is natural to ask whether the shadow obstruction tower of a single class **M** algebra generates the full ideal. Computational evidence at genus 2 and 3 supports this; a proof would require establishing ideal membership in the strata algebra.

10.4. Arithmetic content of shadow denominators. The Virasoro shadow coefficients S_r are rational functions of c with poles at $c = 0$ and $c = -22/5$. The denominators $c^{r-3}(5c + 22)^{\lfloor (r-2)/2 \rfloor}$ grow in a pattern controlled by the Gaussian decomposition. For degrees $r \geq 12$, the numerators may be divisible by Ramanujan-type primes whose appearance is governed by cusp forms. The total depth decomposition $d(\mathcal{A}) = 1 + d_{\text{arith}} + d_{\text{alg}}$ separates the algebraic engine (depths ≤ 4) from the arithmetic frontier (depths ≥ 5).

10.5. General DS primitivity. Theorem 9.2 proves gravitational coproduct primitivity for the principal reduction $\widehat{\mathfrak{sl}}_2 \rightarrow \text{Vir}$. The ghost-number obstruction argument uses only the universal grading of the BRST complex, and should extend to all principal reductions $\widehat{\mathfrak{g}} \rightarrow \mathcal{W}(\mathfrak{g})$ (Conjecture 9.3). Non-principal reductions require a separate analysis: the ghost complex has a finer grading, and the transferred coproduct may be nontrivial.

10.6. The descent fan. The MC element $\Theta_{\mathcal{A}}$ admits three independent projections with distinct targets and distinct symmetries:

- (i) *Categorical.* $\mathcal{A} \rightarrow \bar{B}^{\text{ch}}(\mathcal{A}) \rightarrow \mathcal{A}^!$. The Koszul involution $\mathcal{A} \leftrightarrow \mathcal{A}^!$.
- (ii) *Spectral.* $\mathcal{A} \rightarrow \{S_r\} \rightarrow \sqrt{Q_L} \rightarrow \varepsilon_{Q_L}(s)$. The functional equation $s \leftrightarrow 1 - s$.
- (iii) *Modular.* $\mathcal{A} \rightarrow Z_g(\mathcal{A}) \rightarrow S_{\mathcal{A}}(v)$. The Galois involution on the L -function content.

These three projections share $\Theta_{\mathcal{A}}$ as common source but do not compose into a single functor. The Koszul, functional-equation, and Galois involutions are three different involutions on three different spaces; no functor intertwines all three. For lattice VOAs, all three projections are compatible (the lattice fan is closed). For class **M** algebras, the compatibility is open.

The descent fan is the shadow-tower analogue of the trichotomy between algebraic, analytic, and arithmetic geometry: the categorical projection sees the algebra, the spectral projection sees the analysis, and the modular projection sees the arithmetic. Each is a faithful shadow of $\Theta_{\mathcal{A}}$; none is complete.

10.7. The genus-1 amplitude Mellin transform is Eisenstein. The genus-1 degree-2 shadow amplitude has Fourier coefficients $a_n = -24\kappa \sigma_1(n)$, and the associated Dirichlet series satisfies $D_2(\mathcal{A}, s) = -24\kappa \cdot \zeta(s) \zeta(s-1)$: it is an Eisenstein L -function, not cuspidal. (The shadow L -function $L_{\mathcal{A}}^{\text{sh}}(s) := \sum_{r \geq 2} S_r r^{-s}$, which extracts degree-level constant terms, is a *different* object that does not equal $-\kappa \zeta(s) \zeta(s-1)$.) All connections between the shadow obstruction tower and Riemann zeta zeros are therefore mediated through Eisenstein scattering, not through direct spectral identification. This is both a negative result (the shadow does not “see” individual zeta zeros) and a positive one (the Eisenstein factorisation is a precise structural theorem).

REFERENCES

- [1] A. Beilinson and V. Drinfeld, *Chiral Algebras*, AMS Colloquium Publications, vol. 51, American Mathematical Society, Providence, RI, 2004.
- [2] A. Beilinson, V. Ginzburg, and W. Soergel, Koszul duality patterns in representation theory, *J. Amer. Math. Soc.* **9** (1996), 473–527.
- [3] A. Beilinson and V. Schechtman, Determinant bundles and Virasoro algebras, *Comm. Math. Phys.* **118** (1988), 651–701.
- [4] K. Costello and O. Gwilliam, *Factorization Algebras in Quantum Field Theory*, vols. 1–2, Cambridge University Press, 2017/2021.
- [5] V. Drinfeld and V. Sokolov, Lie algebras and equations of Korteweg–de Vries type, *J. Soviet Math.* **30** (1985), 1975–2036.
- [6] J. Francis and D. Gaitsgory, Chiral Koszul duality, *Selecta Math. (N.S.)* **18** (2012), 27–87.
- [7] C. Faber and R. Pandharipande, Hodge integrals and moduli spaces of curves, in *Moduli of Curves and Abelian Varieties*, Aspects Math. E33, Vieweg, 2000, pp. 109–130.
- [8] B. Feigin and E. Frenkel, Quantization of the Drinfeld–Sokolov reduction, *Phys. Lett. B* **246** (1990), 75–81.
- [9] P. Flajolet and R. Sedgewick, *Analytic Combinatorics*, Cambridge University Press, 2009.
- [10] M. Kontsevich and Yu. Manin, Gromov–Witten classes, quantum cohomology, and enumerative geometry, *Comm. Math. Phys.* **164** (1994), 525–562.
- [11] M. Kontsevich, Deformation quantization of Poisson manifolds, *Lett. Math. Phys.* **66** (2003), 157–216.
- [12] V. Kac, S.-S. Roan, and M. Wakimoto, Quantum reduction for affine superalgebras, *Comm. Math. Phys.* **241** (2003), 307–342.
- [13] J.-L. Loday and B. Vallette, *Algebraic Operads*, Grundlehren der Mathematischen Wissenschaften, vol. 346, Springer, Berlin, 2012.
- [14] D. Mumford, Towards an enumerative geometry of the moduli space of curves, in *Arithmetic and Geometry, Vol. II*, Progr. Math., vol. 36, Birkhäuser, 1983, pp. 271–328.
- [15] S. Priddy, Koszul resolutions, *Trans. Amer. Math. Soc.* **152** (1970), 39–60.
- [16] R. Lorgat, *The Virasoro spectral R-matrix: closed form and non-formality*, preprint, 2026.
- [17] R. Lorgat, *The seven faces of the collision residue*, preprint, 2026.

PERIMETER INSTITUTE FOR THEORETICAL PHYSICS, WATERLOO, ON N2L 2Y5, CANADA

Email address: rlorgat@perimeterinstitute.ca